CRITIC–ADVISER–REVISER CYCLIC REFINEMENT: TOWARDS HIGH-QUALITY EMR CORPUS GENERATION WITH LLMS

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ABSTRACT

Electronic medical records (EMRs) are vital for healthcare research, but their use is limited by privacy concerns. Synthetic EMR generation offers a promising alternative, yet most existing methods merely imitate real records without adhering to rigorous clinical quality principles. To address this, we introduce LLM-CARe, a stage-wise cyclic refinement framework that progressively improves EMR quality through three stages, each targeting a specific granularity: corpus, section and document. At each stage, a Critic, an Adviser, and a Reviser collaborate iteratively to evaluate, provide feedback, and refine the drafts. This structured, multi-stage process produces records that better satisfy clinical quality standards. Experiments show that LLM-CARe significantly enhances EMR quality across all levels compared to strong baselines and yields improved performance on real-world clinical tasks such as diagnosis prediction. Unlike prior work, our method requires no real EMR text for training or prompting, demonstrating the effectiveness of stage-wise, cyclic refinement for generating high-quality, privacy-preserving EMR datasets.

1 INTRODUCTION

Electronic Medical Records (EMRs) are a valuable resource for healthcare research (Ma et al., 2017; Shang et al., 2019; Shen et al., 2025), offering large-scale, clinically grounded insights that reflect real-world medical practice. However, the sensitive nature of patient information poses significant privacy challenges, which severely limit the open sharing and use of real EMRs (Iyengar et al., 2018; Shah & Khan, 2020; Tertulino et al., 2024). To mitigate these concerns, researchers have explored synthetic EMR generation methods that aim to preserve data utility while protecting patient confidentiality (Yan et al., 2022; Murtaza et al., 2023; Yuan et al., 2024; Lin et al., 2025).

Existing EMR synthesis approaches primarily focus on mimicking real records (Lee, 2018; Baowaly et al., 2018; Yoon et al., 2023), without explicit ensuring clinical soundness. This imitation-based strategy is risky: real EMRs may contain errors (Aerts et al., 2021; Mohamed et al., 2023), which can be inadvertently inherited by synthetic data (Figure 1(a)). In practice, EMRs are professional medical documents whose reliability depends on satisfying key requirements such as completeness, consistency, and distribution alignment. Synthetic records that fail to meet such requirements may be less useful—or even misleading—for downstream clinical or research applications.

Recent advances in large language models (LLMs) make them a promising tool for EMR generation, due to their text generation ability and rich internal knowledge. However, as shown in Figure 1(a), our preliminary analysis reveals that LLM outputs often exhibit biased distributions—such as unrealistic gender patterns—and tend to produce only the most typical presentations of a disease, lacking

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评论者–顾问–修订者循环优化：面向使用大型语言模型生成高质量电子病历语料库

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摘要

电子病历（EMR）对医疗研究至关重要，但其使用受到隐私问题的限制。合成电子病历的生成提供了一种有前景的替代方案，然而大多数现有方法仅仅模仿真实病历，而未遵循严格的临床质量原则。为了解决这一问题，我们提出了LLM-CARe，一种分阶段循环精炼框架，通过三个阶段逐步提高EMR质量，每个阶段针对特定的粒度：语料库、章节和文档。在每个阶段中，评论者（Critic）、顾问（Adviser）和修订者（Reviser）协同迭代，对草稿进行评估、提供反馈并进行完善。这一结构化、多阶段的过程生成的病历更能满足临床质量标准。实验表明，与强基线相比，LLM-CARe在各个层面显著提升了EMR质量，并在实际临床任务（如诊断预测）中表现出更好的性能。与以往工作不同，我们的方法不需要真实EMR文本来

1 介绍

电子病历（EMRs）是医疗研究的宝贵资源（Ma 等，2017；Shang 等，2019；Shen 等，2025），提供了大规模、以临床为基础的见解，反映了真实世界的医疗实践。然而，患者信息的敏感性带来了重大的隐私挑战，这严重限制了真实电子病历的公开共享和使用（Iyengar 等，2018；Shah & Khan，2020；Tertulino 等，2024）。为缓解这些问题，研究人员探索了合成电子病历生成方法，旨在在保护患者机密性的同时保持数据的实用性（Yan 等，2022；Murtaza 等，2023；Yuan 等，2024；Lin 等，2025）。

现有的电子病历（EMR）合成方法主要集中在模仿真实病历（Lee, 2018；Baowaly 等, 2018；Yoon 等, 2023），而没有明确确保临床合理性。这种基于模仿的策略存在风险：真实的电子病历可能包含错误（Aerts 等, 2021；Mohamed 等, 2023），这些错误可能会被合成数据无意中继承（图1(a)）。在实践中，电子病历是专业的医疗文档，其可靠性取决于满足完整性、一致性和分布对齐等关键要求。未能满足这些要求的合成病历在下游临床或研究应用中可能用途有限，甚至可能产生误导。

大型语言模型（LLMs）的最新进展使它们成为生成电子病历（EMR）的有前景工具，因为它们具有文本生成能力和丰富的内部知识。然而，如图1(a)所示，我们的初步分析显示，LLM的输出通常表现出偏颇的分布——例如不切实际的性别模式——并且倾向于只生成疾病的最典型表现，缺乏

*通讯作者。

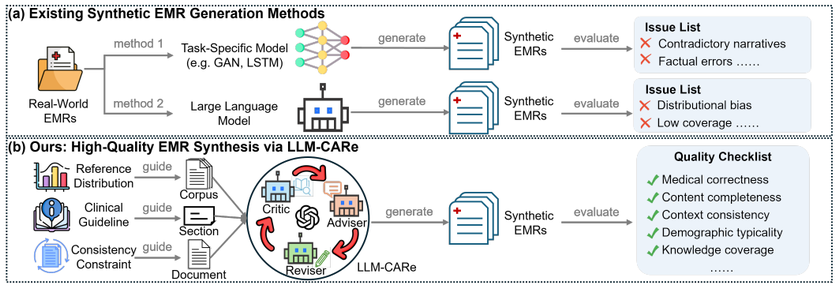


Figure 1: **(a)** Traditional EMR generation that mimic real EMRs without considering quality often leads to suboptimal outputs. **(b)** Our proposed LLM-CARe incorporates cyclic refinement based on quality principles, synthesizing high-quality EMRs.

coverage of the diverse and less common clinical scenarios seen in practice. These challenges highlight the need for more structured approaches to harness LLMs effectively for EMR synthesis.

To bridge this gap, we propose **LLM**-based **C**ritic–**A**dviser–**R**eviser Cyclic **Re**finement (**LLM-CARe**), a framework that enhances synthetic EMR quality through progressive refinement across three levels. As illustrated in Figure 1(b), LLM-CARe incorporates clinical quality principles into the generation process, producing records that align with standards of medical documentation. These requirements are organized into concrete principles of corpus distributional alignment, section completeness, and document consistency, forming the basis for refinement. Guided by them, LLM-CARe proceeds through three stages of refinement—corpus, section, and document—each targeting a distinct aspect of EMR quality. Within every stage, a Critic, an Adviser, and a Reviser collaborate in a cyclic loop: the Critic evaluates the drafts, the Adviser provides targeted feedback, and the Reviser updates the records. While the interaction pattern is shared, the role of each agent adapts to the stage: corpus stage aligns the dataset with realistic distributions, section stage enforces section completeness, and document stage ensures logical consistency within record. This structured process enables systematic enhancement of EMRs from local detail to global corpus characteristics.

To validate the effectiveness of our approach, we conduct experiments on a real-world EMR dataset containing 192k records across 302 diseases. We assess the intrinsic quality of generated records using a strong LLM as a judge, complemented by statistical comparisons to real EMRs. Additionally, we evaluate downstream utility by training models on synthetic EMRs and testing on real records across three representative clinical tasks: diagnosis prediction, examination recommendation, and treatment recommendation. Results show that LLM-CARe consistently outperforms baselines in both record quality and task performance. Notably, our method requires no access to real EMR text for training or prompting during generation, fully preserving patient privacy while producing clinically meaningful and practically useful EMRs. Our main contributions are summarized as follows:

- We propose **LLM-CARe**, a stage-wise multi-agent framework for high-quality EMR synthesis that employs cyclic refinement based on clinical quality principles.
- LLM-CARe consistently improves EMR quality compared to baselines across multiple levels. Further analysis shows that all three agents play essential and complementary roles.
- Without training or prompting on real EMR text, our synthetic data yields superior downstream task performance compared to baselines, ensuring both utility and privacy.

2 Related Work

There has been growing interest in synthesizing EMRs to address privacy concerns and facilitate secure data sharing. We categorize existing methods into three main paradigms:

GAN-based EMR Generation. Generative adversarial networks (GANs) have been extensively explored for EMR synthesis. Some works generate EMRs from random noise vectors (Choi et al.,

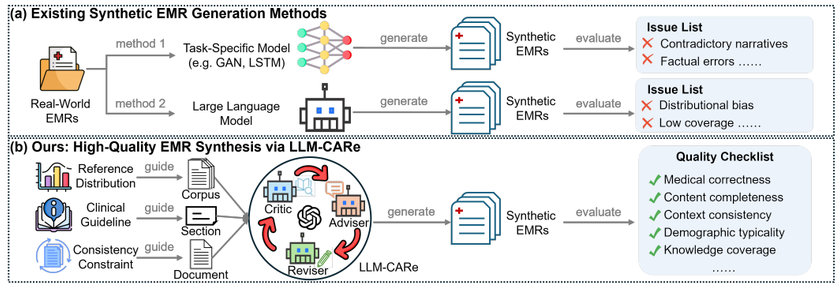


图1：（a）传统的电子病历（EMR）生成模拟真实EMR，但不考虑质量，往往导致次优输出。（b）我们提出的LLM-CARe基于质量原则重新引入循环优化，合成高质量的EMR。

涵盖实践中遇到的多样化和较不常见的临床场景。这些挑战突显了更有结构的方法以有效利用大型语言模型进行电子病历综合的必要性。

为了弥合这一差距，我们提出了基于大语言模型的批评者–顾问–修订者循环改进（LLM-CARe）框架，通过在三个层面的逐步改进来提升合成电子病历（EMR）的质量。如图1(b)所示，LLM-CARe将临床质量原则融入生成过程，生成符合医疗文档标准的病历。这些要求被组织为具体的语料库分布对齐、章节完整性和文档一致性原则，构成改进的基础。在这些原则的指导下，LLM-CARe通过三个改进阶段进行——语料库、章节和文档，每个阶段针对EMR质量的不同方面。在每个阶段内，批评者、顾问和修订者在循环中协作：批评者评估草稿，顾问提供针对性反馈，修订者更新病历。虽然交互模式相同，但每个角色在不同阶段的作用会有所调整：语料库阶段对齐

为了验证我们方法的有效性，我们在一个包含192,000条记录、涉及302种疾病的真实电子病历（EMR）数据集上进行了实验。我们通过使用强大的大语言模型（LLM）作为评判来评估生成记录的内在质量，并辅以与真实电子病历的统计比较。此外，我们通过在合成电子病历上训练模型并在三种代表性临床任务上使用真实记录进行测试来评估下游效用，这三种任务分别是：诊断预测、检查推荐和治疗推荐。结果表明，LLM-CARe在记录质量和任务性能上均持续优于基线方法。值得注意的是，我们的方法在生成过程中无需访问真实电子病历文本，从而完全保护了患者隐私，同时生成临床上有意义且实际有用的电子病历。我们的主要贡献总结如下：

- 我们提出了LLM-CARe，一种分阶段的多代理高质量电子病历合成框架，该框架基于临床质量原则采用循环优化。
- LLM-CARe在多个层面上相比基线方法持续提高了电子病历质量。进一步分析表明，所有三个代理都扮演着重要且互补的角色。
- 在没有对真实电子病历（EMR）文本进行训练或提示的情况下，我们的合成数据在下游任务中的表现优于基线，确保了实用性和隐私性。

2 相关工作

The人们对合成电子病历以解决隐私问题并促进……的兴趣日益增加 e
secu关于数据共享。我们将现有的方法分为三种主要范式：

基于GAN的电子病历生成。生成对抗网络（GANs）已被广泛用于电子病历合成。一些研究工作从随机噪声向量生成电子病历（Choi 等，

2017; Baowaly et al., 2018; Chin-Cheong et al., 2019; Yoon et al., 2023), while other methods incorporate structured conditions—such as diagnosis codes—to guide generation process (Rashidian et al., 2020; Zhang et al., 2020; Guan et al., 2021; Li et al., 2023). Although effective at modeling data distributions, these methods typically ignore the clinical quality of the generated records.

Auto-regressive EMR Generation. Another line of research leverages auto-regressive models to generate EMRs. Recurrent neural networks (RNNs) have been used to model sequential EMR data (Lee, 2018; Melamud & Shivade, 2019; Mosquera et al., 2023; Ganguli et al., 2023), and more recently, transformer-based architectures have been introduced to capture long-range dependencies within records (Wang et al., 2019; Amin-Nejad et al., 2020; Theodorou et al., 2023; Karami et al., 2024). While these models excel at learning temporal and structural patterns, they generally treat EMRs as sequences of tokens without mechanisms to ensure clinically meaningful coherence.

LLM-based EMR Generation. With the emergence of large language models (LLMs), recent studies have explored prompting LLMs to synthesize EMRs, either by providing brief clinical descriptions or by asking the model to emulate real patient records (Litake et al., 2024; Kumichev et al., 2024; Abdel-Khalek et al., 2024; Kweon et al., 2024; Lin et al., 2025). While LLMs exhibit strong capabilities, direct generation often results in diverge from realistic corpus-level distributions.

Recent work has also explored diffusion models for EMR synthesis (Yuan et al., 2023), but existing methods primarily target numerical representations rather than free-text EMRs, making them less applicable to the textual EMR generation setting studied in this work.

3 METHOD

In this section, we present **LLM-CARe**, a stage-wise cyclic refinement framework for enhancing the quality of synthetic EMRs. Unlike ordinary free-form text, EMR must satisfy quality requirements to be clinically meaningful. We focus on five principles across three levels: **demographic typicality** (plausible population patterns), **knowledge coverage** (sufficient breadth of clinical conditions), **content completeness** (each section includes its essential information), **medical correctness** (information is clinically valid), and **context consistency** (sections do not contradict one another). Each principle is further divided into concrete criteria that can be explicitly checked and refined. Figure 2 shows representative examples, with full details in Appendix A.

Level	Principle	Example of Criteria
Corpus	Demographic Typicality	The gender ratio of synthetic data should align with real populations.
	Knowledge Coverage	The synthetic dataset should cover comprehensive symptoms.
Section	Content Completeness	The chief complaint should state the reason for admission.
Document	Medical Correctness	The diagnosis should be valid given the patient's gender.
	Context Consistency	Symptoms in chief complaint and history of present illness should align.

Figure 2: Quality principles across three levels with representative criteria.

To address these requirements, LLM-CARe refines EMRs in three successive stages—corpus, section, and document. As illustrated in Figure 3, each stage involves the collaboration of three agents in a cyclic loop: the *Critic*, who evaluates drafts against stage-specific objectives; the *Adviser*, who pinpoints areas for improvement and suggests strategies; and the *Reviser*, who incorporates feedback to update the drafts. This iterative process progressively improves EMRs from global distributional alignment, to field-level completeness, and finally to record-level coherence.

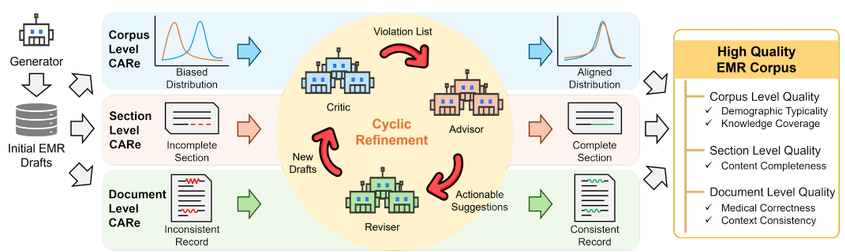


Figure 3: Overview of our proposed LLM-CARe framework for synthesizing high-quality EMRs.

2017；Baowaly 等，2018；Chin-Cheong 等，2019；Yoon 等，2023），而其他方法则融合结构化条件——例如诊断代码——以指导生成过程（Rashidian 等，2020；Zhang 等，2020；Guan 等，2021；Li 等，2023）。尽管在建模数据分布方面有效，这些方法通常忽略生成记录的临床质量。

自回归电子病历生成。另一种研究方向利用自回归模型生成电子病历。循环神经网络（RNNs）已被用于建模序列化的电子病历数据（Lee, 2018；Melamud & Shivade, 2019；Mosquera 等, 2023；Ganguli 等, 2023），而最近，基于Transformer的架构被引入以捕捉记录中的长程依赖（Wang 等, 2019；Amin-Nejad 等, 2020；Theodorou 等, 2023；Karami 等, 2024）。虽然这些模型在学习时间和结构模式方面表现出色，但它们通常将电子病历视为令牌序列，缺少确保临床意义一致性的机制。

基于大语言模型的电子病历生成。随着大语言模型（LLMs）的出现，近期的研究探索了通过提示LLMs来合成电子病历的方法，或者提供简要的临床描述，或者要求模型模拟真实的患者记录（Litake 等，2024；Kumichev 等，2024；Abdel-Khalek 等，2024；Kweon 等，2024；Lin 等，2025）。尽管LLMs表现出强大的能力，但直接生成往往会偏离现实的语料库层分布。

最近的工作也探索了用于电子病历合成的扩散模型（Yuan 等，2023），但现有方法主要针对数值表示，而非自由文本的电子病历，使其在本研究中所研究的文本电子病历生成场景中应用性较低。

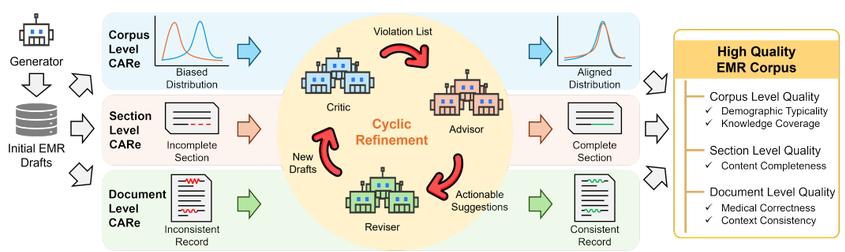
3 方法

在本节中，我们介绍了 LLM-CARe，一种分阶段循环改进框架，用于提升合成电子病历（EMR）的质量。与普通的自由文本不同，电子病历必须满足质量要求才能具有临床意义。我们关注三个层面的五项原则：人口统计典型性（合理的人群模式）、知识覆盖面（临床状况的充分广度）、内容完整性（每个部分包含其必要信息）、医学正确性（信息在临床上有效）以及上下文一致性（各部分之间不互相矛盾）。每项原则进一步分成可以明确检查和改进的具体标准。图2显示了代表性示例，详细信息见附录A。

Level	Principle	Example of Criteria
Corpus	Demographic Typicality	The gender ratio of synthetic data should align with real populations.
	Knowledge Coverage	The synthetic dataset should cover comprehensive symptoms.
Section	Content Completeness	The chief complaint should state the reason for admission.
Document	Medical Correctness	The diagnosis should be valid given the patient's gender.
	Context Consistency	Symptoms in chief complaint and history of present illness should align.

图 2：跨三个层级的质量原则及其代表性标准。

为了满足这些需求，LLM-CARe 在三个连续阶段对电子病历（EMR）进行优化——语料库、章节和文档。如图 3 所示，每个阶段都涉及三个智能体在循环中协作：批评者（Critic），根据阶段特定目标评估草稿；顾问（Adviser），指出需要改进的地方并提出策略建议；修订者（Reviser），根据反馈更新草稿。这个迭代过程逐步提升电子病历，从整体分布对齐，到字段级完整性，最终达到记录级连贯性。



Fig图3：我们提出的用于合成高质量电子病历的LLM-CARe框架概述 .

3.1 INITIAL DRAFT GENERATION

The generation process begins with a *generator* agent $\mathcal{M}_{\text{generator}}$, which produces initial EMR drafts based on an input prompt x. This prompt specifies key information such as the target primary diagnosis and required EMR fields (e.g., chief complaint, history of present illness). For each prompt, the generator samples multiple drafts to form a starting draft pool:

$$\mathcal{D}^{(0)} = \{E_1^{(0)}, \ldots, E_n^{(0)}\}, \quad E_i^{(0)} \sim \mathcal{M}_{\text{generator}}(x) \tag{1}$$

where $\mathcal{D}^{(0)}$ denotes the initial draft set and each $E_i^{(0)}$ is an EMR instance. These drafts may exhibit omissions, inconsistencies, or clinically implausible details, but they provide the foundation for subsequent stage-wise refinement.

At the dataset scale, high-quality synthetic EMRs must preserve realistic and representative distributions. We focus on two aspects: **Demographic Typicality**, ensuring variables such as age and gender reflect real-world patient populations (see Figure 11 for examples), and **Knowledge Coverage**, ensuring the corpus contains both common and rare clinical conditions (with detailed dimensions listed in Table 9). Maintaining these corpus-level properties is essential, as deviations may introduce demographic bias or insufficient coverage of clinically important but less frequent scenarios. These goals are addressed through corpus-level agent interactions, where the critic, adviser, and reviser collaborate to align the overall distribution.

Corpus-level Critic. At the corpus stage, the critic focuses on dataset-wide properties, capturing how well synthetic EMRs aligns with target distributions. For each attribute $c_{\text{corpus},k}$ (e.g., age, gender, or a knowledge dimension), it measures the deviation of the current corpus $\mathcal{D}^{(t)}$ from the reference distribution $\mathcal{T}_d$ derived from aggregate statistics on the training set, including age and gender distributions and the frequencies of clinical concepts of interest (e.g., symptoms, medications):

$$\delta_{\text{corpus},k}^{(t)} = \mathcal{M}_{\text{critic}}^{\text{corpus}}\left(\mathcal{D}^{(t)}, \mathcal{T}_d, c_{\text{corpus},k}\right) \tag{2}$$

Corpus-level Adviser. The adviser interprets the critic's feedback to guide modifications at the dataset level. Based on the deviations, it identifies a subset of records $\mathcal{S}_k^{(t)} \subset \mathcal{D}^{(t)}$ whose adjustment would most effectively reduce distributional mismatch, and generates actionable feedback $F_{\text{corpus},k}^{(t)}$:

$$\mathcal{S}_k^{(t)}, F_{\text{corpus},k}^{(t)} = \mathcal{M}_{\text{adviser}}^{\text{corpus}}\left(\mathcal{D}^{(t)}, c_{\text{corpus},k}, \delta_{\text{corpus},k}^{(t)}\right) \tag{3}$$

Corpus-level Reviser. The reviser applies the adviser's instructions to the selected subset $\mathcal{S}_k^{(t)}$, modifying or enriching records to better match the reference distribution:

$$\mathcal{S}_k^{(t+1)} = \mathcal{M}_{\text{reviser}}^{\text{corpus}}\left(\mathcal{S}_k^{(t)}, F_{\text{corpus},k}^{(t)}\right) \tag{4}$$

3.2 SECTION-LEVEL CARE

At the section scale, high-quality EMRs must ensure **Content Completeness**: each field should contain the essential clinical elements expected for its type. Ensuring section-level completeness is crucial, as omissions in key fields can result in records that misrepresent the patient's condition and compromise clinical validity. To operationalize this, we define a set of section-specific criteria derived from clinical guidelines (see Table 6), and apply cyclic refinement with critic, adviser, and reviser agents to supplement missing information.

Section-level Critic. The critic operates for each single section. For a section $s_{i,m}^{(t)}$ in record $E_i^{(t)}$ and a criterion $c_{\text{sec},k}$ derived from clinical guidelines, it determines whether the criterion is met:

$$\delta_{\text{sec},i,k}^{(t)} = \mathcal{M}_{\text{critic}}^{\text{sec}}\left(s_{i,m}^{(t)}, c_{\text{sec},k}\right), \quad \delta_{\text{sec},i,k}^{(t)} \in \{0,1\} \tag{5}$$

Section-level Adviser. For unmet criteria ($\delta_{\text{sec},i,k}^{(t)} = 0$), the adviser examines the section and designs specific instructions to indicate exactly which clinical elements should be added or clarified:

$$F_{\text{sec},i,k}^{(t)} = \mathcal{M}_{\text{adviser}}^{\text{sec}}\left(s_{i,m}^{(t)}, c_{\text{sec},k}\right) \tag{6}$$

3.1 初稿生成

生成过程从生成器代理 $\mathcal{M}_{\text{generator}}$ 开始，该代理根据输入提示 x 生成初步的电子病历草稿。该提示指定了关键信息，如目标主要诊断和所需的电子病历字段（例如，主诉、现病史）。对于每个提示，生成器会抽样多个草稿以形成起始草稿池：

$$\mathcal{D}^{(0)} = \{E_1^{(0)}, \ldots, E_n^{(0)}\}, \quad E_i^{(0)} \sim \mathcal{M}_{\text{generator}}(x) \tag{1}$$

其中 $\mathcal{D}^{(0)}$ 表示初始草稿集合，每个 $E_i^{(0)}$ 是一个电子病历（EMR）实例。这些草稿可能存在遗漏、不一致或临床上不合理的细节，但它们为后续分阶段的完善提供了基础。

在数据集规模上，高质量的合成电子病历必须保持真实且具有代表性的分布。我们关注两个方面：人口特征典型性，确保年龄和性别等变量反映真实世界的患者群体（示例见图 11）；以及知识覆盖，确保语料库包含常见和罕见的临床状况（详细维度见表 9）。保持这些语料库级别的特性至关重要，因为偏离可能会引入人口统计学偏差或临床上重要但不常见的情景覆盖不足。这些目标通过语料库级别的代理交互来实现，其中评论者、顾问和修订者协作以使整体分布保持一致。

语料库级别的评审。在语料库阶段，评审者关注数据集的整体属性，评估合成电子病历（EMR）与目标分布的匹配程度。对于每个属性$c_{\text{corpus},k}$（例如年龄、性别或某个知识维度），它衡量当前语料库 $\mathcal{D}^{(t)}$ 与从训练集聚合统计中得出的参考分布 $\mathcal{T}_d$ 的偏差，包括年龄和性别分布，以及感兴趣的临床概念的频率（例如症状、药物）：

$$\delta_{\text{corpus},k}^{(t)} = \mathcal{M}_{\text{critic}}^{\text{corpus}}\left(\mathcal{D}^{(t)}, \mathcal{T}_d, c_{\text{corpus},k}\right) \tag{2}$$

语料库级顾问。该顾问解释评论者的反馈，以指导数据集层面的修改。根据偏差，它识别出一部分记录 $\mathcal{S}_k^{(t)} \subset \mathcal{D}^{(t)}$，调整这些记录可以最有效地减少分布不匹配，并生成可操作的反馈 $F^{(t)}$ 语料库 $_k$:

$$\mathcal{S}_k^{(t)}, F_{\text{corpus},k}^{(t)} = \mathcal{M}_{\text{adviser}}^{\text{corpus}}\left(\mathcal{D}^{(t)}, c_{\text{corpus},k}, \delta_{\text{corpus},k}^{(t)}\right) \tag{3}$$

语料库级修订器。修订器将顾问的指示应用于选定的子集 $\mathcal{S}_k^{(t)}$，修改或丰富记录以更好地匹配参考分布：

$$\mathcal{S}_k^{(t+1)} = \mathcal{M}_{\text{reviser}}^{\text{corpus}}\left(\mathcal{S}_k^{(t)}, F_{\text{corpus},k}^{(t)}\right) \tag{4}$$

3.2 节级护理

在章节层面，高质量的电子病历必须确保内容完整性：每个字段都应包含其类型所期望的关键临床元素。确保章节级别的完整性至关重要，因为关键字段的遗漏可能导致记录不能准确反映患者的状况，从而影响临床有效性。为了实现这一点，我们定义了一组源自临床指南的章节特定标准（见表6），并通过批评者、顾问和修改者代理进行循环优化，以补充缺失的信息。

章节级评审。评审针对每个单独的章节进行。对于记录 $E_i^{(t)}$ 中的章节 $s_i^{(t)}$,$_m$ 以及根据临床指南得出的标准 $c_{\text{sec},k}$，它确定该标准是否被满足：

$$\delta_{\text{sec},i,k}^{(t)} = \mathcal{M}_{\text{critic}}^{\text{sec}}\left(s_{i,m}^{(t)}, c_{\text{sec},k}\right), \quad \delta_{\text{sec},i,k}^{(t)} \in \{0,1\} \tag{5}$$

章节级顾问。对于未满足的标准（$\delta^{(t)}$ 节 $_{i,k} = 0$），顾问会检查该章节并设计具体说明，以明确指出应添加或澄清的临床要素：

$$F_{\text{sec},i,k}^{(t)} = \mathcal{M}_{\text{adviser}}^{\text{sec}}\left(s_{i,m}^{(t)}, c_{\text{sec},k}\right) \tag{6}$$

Section-level Reviser. Using the adviser's guidance, the reviser updates the section by incorporating the recommended elements while preserving existing content and coherence:

$$s_{i,m}^{(t+1)} = \mathcal{M}_{\text{reviser}}^{\text{sec}}\left(s_{i,m}^{(t)}, F_{\text{sec},i,k}^{(t)}\right) \tag{7}$$

Through this cycle, sections are progressively completed and made sufficient for their clinical role.

3.3 Document-Level CARe

At the document scale, EMRs must ensure both **Medical Correctness**—that clinical statements are valid given the diagnosis—and **Context Consistency**—that information across sections does not conflict. Without enforcing these criteria, even small cross-section inconsistencies or incorrect clinical relations can undermine the internal logic of the record to the point where the overall EMR becomes clinically unreliable. To make these requirements concrete, we define detailed criteria for both aspects (see Tables 7 and Table 8). To enforce them, we refine EMRs through document-level agent interactions, where the focus is on coherence across multiple sections.

Document-level Critic. The critic evaluates each record as a whole, checking constraints across sections for logical and clinical plausibility. For a consistency criterion $c_{\text{doc},k}$, it outputs a judgment:

$$\delta_{\text{doc},i,k}^{(t)} = \mathcal{M}_{\text{critic}}^{\text{doc}}\left(E_i^{(t)}, c_{\text{doc},k}\right), \quad \delta_{\text{doc},i,k}^{(t)} \in \{0,1\} \tag{8}$$

Document-level Adviser. When inconsistencies are flagged ($\delta_{\text{doc},i,k}^{(t)} = 0$), the adviser generates targeted feedback, often suggesting edits to the less influential section to restore harmony:

$$F_{\text{doc},i,k}^{(t)} = \mathcal{M}_{\text{adviser}}^{\text{doc}}\left(E_i^{(t)}, c_{\text{doc},k}\right) \tag{9}$$

Document-level Reviser. Finally, the reviser integrates this feedback to harmonize the conflicting sections and yield an updated document:

$$E_i^{(t+1)} = \mathcal{M}_{\text{reviser}}^{\text{doc}}\left(E_i^{(t)}, F_{\text{doc},i,k}^{(t)}\right) \tag{10}$$

Through this process, records are refined into coherent, consistent, and clinically valid narratives.

3.4 Stage-wise Ordering

At each stage, the critic, adviser, and reviser interact in cycles to refine the drafts according to stage-specific principles. Once the drafts have been improved at the current granularity, they are passed to the next stage, where the agent interaction continues under a different focus. The staged order is intentional: modifications at one level can influence others, so we proceed **from the most flexible to the most stringent stage**. Corpus-level refinement is relatively soft, aiming to align distributions without requiring exact matches, and is therefore performed first. While document-level refinement enforces strict logical consistency across sections, where errors could introduce serious contradictions, thus is performed last. By progressing in this order, each stage builds on the previous one while minimizing unintended conflicts. Through this staged refinement, the synthetic EMRs achieve high quality across corpus, section, and document levels.

4 Experimental Setup

Dataset To validate the effectiveness of our method, we conduct experiments on a real-world EMR dataset comprising 192k records across 302 diseases. The dataset is carefully de-identified by removing all sensitive patient information. Unlike many prior studies that focus on synthesizing a single field in isolation (e.g., chief complaint), we consider multiple fields that together capture the clinical trajectory from admission to discharge to provide a comprehensive view of each clinical episode. To ensure consistent disease distribution across subsets, we perform an 8:2 stratified split based on disease categories, maintaining proportional representation in both the training and test sets. Further details are provided in Appendix B.

章节级修改者。使用顾问的指导，修改者通过纳入推荐的元素来更新章节，同时保留现有内容和连贯性：

$$s_{i,m}^{(t+1)} = \mathcal{M}_{\text{reviser}}^{\text{sec}}\left(s_{i,m}^{(t)}, F_{\text{sec},i,k}^{(t)}\right) \tag{7}$$

Thr在这个周期中，各部分逐步完成并达到其临床作用所需的充分程度 .

3.3 文档级别的护理

在文档层面上，电子病历（EMRs）必须确保医疗正确性——即在给定诊断的情况下临床陈述是有效的——以及上下文一致性——即各部分信息之间不冲突。如果不执行这些标准，即使是小的跨部分不一致或错误的临床关系，也可能破坏病历的内部逻辑，导致整体电子病历在临床上不可靠。为了使这些要求具体化，我们为两个方面定义了详细标准（见表7和表8）。为了执行这些标准，我们通过文档级代理交互来优化电子病历，重点是多部分之间的连贯性。

文档级审查员。审查员对每条记录作为整体进行评估，检查各部分之间的约束条件，以确保逻辑和临床上的合理性。对于一致性标准$c_{\text{doc},k}$，它会输出一个判断结果：

$$\delta_{\text{doc},i,k}^{(t)} = \mathcal{M}_{\text{critic}}^{\text{doc}}\left(E_i^{(t)}, c_{\text{doc},k}\right), \quad \delta_{\text{doc},i,k}^{(t)} \in \{0,1\} \tag{8}$$

文档级顾问。当发现不一致之处时（$\delta^{(t)}$ 文档 $_{i,k} = 0$），顾问会生成针对性的反馈，通常建议对影响较小的部分进行修改以恢复协调：

$$F_{\text{doc},i,k}^{(t)} = \mathcal{M}_{\text{adviser}}^{\text{doc}}\left(E_i^{(t)}, c_{\text{doc},k}\right) \tag{9}$$

文档级修订者。最后，修订者整合这些反馈，以协调冲突的部分并生成更新的文档：

$$E_i^{(t+1)} = \mathcal{M}_{\text{reviser}}^{\text{doc}}\left(E_i^{(t)}, F_{\text{doc},i,k}^{(t)}\right) \tag{10}$$

通过这一过程，记录被提炼成连贯、一致且临床上有效的叙述。

3.4 分阶段订购

在每个阶段，批评者、顾问和修改者以循环方式互动，根据特定阶段的原则优化草稿。一旦草稿在当前粒度上得到改进，它们会被传递到下一阶段，在那里代理的互动在不同的重点下继续进行。阶段顺序是有意安排的：一个层面的修改可能会影响其他层面，因此我们从最灵活的阶段开始，逐步过渡到最严格的阶段。语料层面的精炼相对宽松，旨在对齐分布而不要求完全匹配，因此首先进行。而文档层面的精炼则在各部分之间强制严格的逻辑一致性，因为错误可能引入严重的矛盾，因此最后进行。通过按此顺序推进，每个阶段在前一阶段的基础上进行，同时将非预期冲突降到最低。通过这种分阶段的精炼，合成电子病历在语料、章节和文档层面都能达到高质量。

4 实验设置

数据集 为了验证我们方法的有效性，我们在一个真实的电子病历（EMR）数据集上进行了实验，该数据集包含192,000条记录，涵盖302种疾病。通过移除所有敏感患者信息，数据集已被仔细去标识化。不同于许多先前研究只关注单个字段的合成（例如主诉），我们考虑了多个字段，这些字段共同捕捉了从入院到出院的临床轨迹，从而提供每个临床病例的全面视图。为了确保各子集之间的疾病分布一致，我们基于疾病类别进行了8:2的分层划分，在训练集和测试集中保持了比例代表性。更多细节请参见附录B。

Baselines We compare our method against two categories of baselines. **(1) Non-LLM generative models. LSTM** (Lee, 2018) and **mtGAN** (Guan et al., 2021) share the same generation setup—both take the disease label as input and are trained from scratch on the training split to produce full EMRs, differing only in model architecture (LSTM vs. GAN). **(2) LLM-based generators. MedSyn** (Kumichev et al., 2024) generates EMRs by prompting an LLM with real EMR examples from the training set along with realted symptoms; **LLM Direct** generates EMRs from instructions without quality control; and **Self-Refine** (Madaan et al., 2023) performs iterative self-critique that evaluates all quality requirements jointly in each refinement round. None of these LLM-based baselines involve fine-tuning. All approaches are conditioned on disease labels, and generate the same number of EMRs with same disease distribution as the test set. Details are provided in Appendix C.

Evaluation Settings We employ two types of metrics to evaluate synthetic EMRs:

EMR quality is assessed based on the five principles introduced above. For medical correctness, content completeness, and context consistency, we use an LLM-based judge that evaluates each EMR against our criterion set. For demographic typicality and knowledge coverage, we compute statistical similarity between synthetic and real data within each disease—comparing age and gender distributions, and measuring the proportion of clinically relevant entities (symptoms, exams, treatments) covered in synthetic EMRs.

Downstream task performance provides an practical way to evaluate the utility of synthetic data. Following prior work, we train task-specific models on synthetic EMRs and evaluate their performance on real-world test data. This setup reflects a common use case of synthetic data in low-resource scenarios. We consider three representative tasks of high clinical relevance: diagnosis prediction, examination recommendation, and treatment recommendation. These tasks collectively span key aspects of medical decision-making. Each task is framed as a multiple-choice question, where the model predicts answers based on the chief complaint and history of present illness. Illustrative examples are shown in Figure 4. More details are provided in Appendix D.

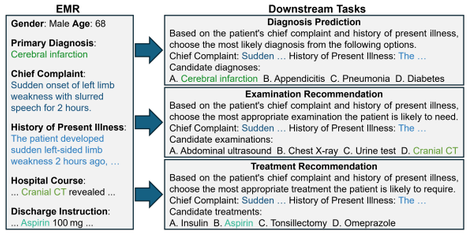


Figure 4: Construction of multiple-choice questions for three downstream tasks from an EMR.

Implementation Details We use Qwen2.5-7B-Instruct (Yang et al., 2025) as the backbone model for all LLM-based methods. For EMR quality evaluation, we adopt Qwen2.5-32B-Instruct, as larger models tend to provide more reliable judgments. For downstream tasks, we fine-tune Qwen2.5-0.5B-Instruct on synthetic EMRs and evaluate performance on real test data. Detailed experimental settings are provided in Appendix E.

5 Experimental Results and Discussion

5.1 Comparison of EMR Quality

Table 1 summarizes performance across the five quality principles. LLM-CARe consistently outperforms all baselines on every metric, improving both individual-record quality and corpus-level characteristics. Among the baselines, LLM-based baselines generally exceed traditional models on section- and document-level criteria, yet without structured refinement they can underperform on corpus-level dimensions. Self-Refine provides only minor gains over LLM Direct and remains far below our method, indicating that generic all-in-one revision cannot satisfy EMRs' multi-level quality requirements. This shows that **simply employing LLMs is not sufficient to guarantee comprehensive EMR quality**. In contrast, LLM-CARe achieves strong and balanced improvements across all levels through principle-guided, stage-wise refinement. To confirm that these findings are not tied to a specific judge, we also evaluate using additional LLMs (Appendix F). Fine-grained subgroup analyses across demographic and disease-frequency partitions are provided in Figure 12.

基线 我们将我们的方法与两类基线进行比较。 (1) 非LLM生成模型。LSTM (Lee, 2018) 和 mtGAN (Guan et al., 2021) 共享相同的生成设置——两者都以疾病标签为输入，并从训练集从头训练以生成完整的电子病历，只是在模型架构上有所不同（LSTM 与 GAN）。 (2) 基于LLM的生成器。MedSyn (Kumichev et al., 2024) 通过使用训练集中的真实电子病历示例及相关症状提示LLM生成电子病历；LLM Direct 从指令生成电子病历，但没有质量控制；Self-Refine (Madaan et al., 2023) 执行迭代自我批评，在每轮优化中联合评估所有质量要求。这些基于LLM的基线方法都不涉及微调。所有方法都以疾病标签为条件，生成的电子病历数量及疾病分布与测试集相同。详细信息见附录 C。

EvaluationSettings 我们使用两种类型的指标来评估合成电子病历（EMR）：

电子病历（EMR）的质量评估基于上述五个原则。对于医疗正确性、内容完整性和上下文一致性，我们使用基于大型语言模型（LLM）的评估器，根据我们的标准集合对每份电子病历进行评估。对于人口统计典型性和知识覆盖率，我们在每种疾病内计算合成数据与真实数据的统计相似性——比较年龄和性别分布，并测量合成电子病历中涉及的临床相关实体（症状、检查、治疗）的覆盖比例。

下游任务性能提供了一种评估合成数据实用性的实际方法。按照以往的研究，我们在合成电子病历（EMRs）上训练特定任务的模型，并在真实世界的测试数据上评估其性能。这种设置反映了合成数据在资源匮乏场景中的常见使用情况。我们考虑三种具有高临床相关性的代表性任务：诊断预测、检查推荐和治疗推荐。这些任务共同涵盖了医疗决策的关键方面。每个任务都被设计为多项选择问题

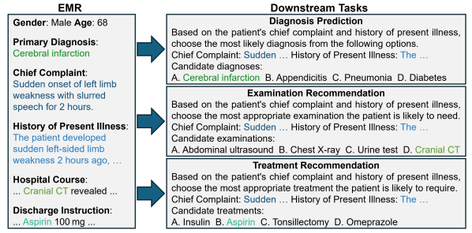


图4：从电子病历构建三个下游任务的多项选择题。

tion，其中模型根据主要症状和现病史预测答案。示例见图4。更多细节见附录D。

实现细节 我们使用 Qwen2.5-7B-Instruct（Yang 等，2025）作为所有基于大语言模型的方法的基础模型。对于电子病历质量评估，我们采用 Qwen2.5-32B-Instruct，因为较大的模型往往能提供更可靠的判断。对于下游任务，我们在合成电子病历上微调 Qwen2.5-0.5B-Instruct，并在真实测试数据上评估性能。详细的实验设置见附录 E。

5 E实验结果与讨论 SION

5.1 电子病历质量的比较

表 1 总结了五项质量原则下的表现。LLM-CARe 在每个指标上始终优于所有基线模型，同时提升了单条记录的质量和语料库级别的特性。在基线模型中，基于 LLM 的方法通常在章节和文档级指标上超过传统模型，但如果没有结构化的优化，它们在语料库级维度上可能表现不佳。Self-Refine 相较于 LLM Direct 仅提供了微小的提升，且远低于我们的方法，这表明通用的一体化修订无法满足电子病历的多级质量需求。这表明，仅仅使用 LLM 并不足以保证全面的电子病历质量。相比之下，LLM-CARe 通过以原则为指导、分阶段的优化，在各个层面都取得了强劲且均衡的提升。为了确认这些发现并不依赖于特定的评审者，我们还使用了额外的 LLM 进行了评估（附录 F）。针对人口统计和其他因素的细粒度子群分析显示

Table 1: Quality score (%) of generated EMRs across principles, where higher values indicate better performance. (*) denotes standard deviation calculated from 3 runs with different random seeds.

Type	Method	Rely on EMR Text	Section Level	Document Level		Corpus Level	
			Content Completeness	Medical Correctness	Context Consistency	Demographic Typicality	Knowledge Coverage
Non-LLM	LSTM	✓	70.8(1.1)	65.0(0.4)	21.7(2.3)	93.3(0.6)	70.4(0.4)
	mtGAN	✓	55.8(2.9)	51.8(6.2)	21.4(4.2)	93.6(1.4)	76.3(3.7)
LLM-Based	MedSyn	✓	84.8(0.3)	95.3(0.8)	91.9(1.1)	84.1(0.9)	84.5(5.8)
	LLM Direct	×	77.1(0.1)	90.7(0.2)	87.9(0.1)	77.7(0.1)	73.9(0.2)
	Self-Refine	×	78.3(0.2)	90.9(0.0)	88.5(0.1)	77.7(0.4)	78.0(2.4)
	LLM-CARe(ours)	×	**91.2**(0.4)	**98.6**(0.0)	**93.8**(0.1)	**96.8**(1.4)	**94.1**(0.1)

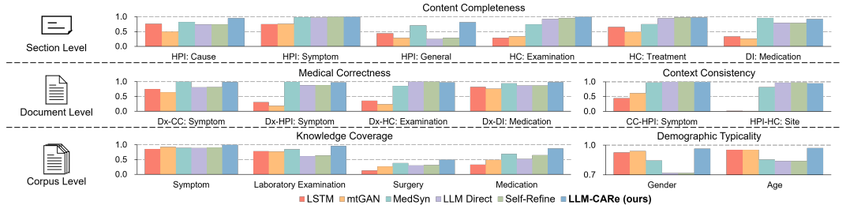


Figure 5: Detailed EMR quality evaluation across 3 levels. Abbreviations: CC-Chief Complaint, HPI-History of Present Illness, HC-Hospital Course, DI-Discharge Instructions, Dx-Diagnosis.

Figure 5 presents a fine-grained breakdown of EMR quality across several representative and clinically important criteria; complete results are provided in Appendix H. Our method achieves the best or competitive performance on the majority of criteria, demonstrating robustness across diverse quality dimensions. Notably, MedSyn underperforms even the LLM Direct on some criteria, such as the completeness of the hospital course. **This suggests that real EMRs—used by MedSyn as in-context exemplars—may contain omissions that propagate into the generated records.** These findings further highlight the limitations of purely imitative approaches and emphasize the importance of explicitly modeling and enforcing quality standards during generation.

5.2 Comparison of Downstream Task Performance

Table 2: Accuracy (%) of three downstream tasks, where micro and macro are averaged across diseases. (*) denotes standard deviation calculated from 3 runs with different random seeds.

Type	Method	Rely on EMR Text	Diagnosis Prediction		Examination Recommendation		Treatment Recommendation	
			Micro	Macro	Micro	Macro	Micro	Macro
Non-LLM	LSTM	✓	74.0(2.0)	73.1(2.0)	75.7(0.3)	76.4(0.2)	56.7(0.6)	50.0(0.7)
	mtGAN	✓	81.9(2.2)	80.9(2.5)	72.4(1.5)	73.4(1.4)	58.6(2.8)	52.9(3.0)
LLM-Based	MedSyn	✓	81.7(0.0)	81.7(0.0)	82.9(0.1)	82.2(0.1)	74.5(0.1)	71.3(0.2)
	LLM Direct	×	81.8(0.0)	81.8(0.2)	64.4(0.0)	65.4(0.0)	60.9(0.2)	59.0(0.3)
	Self-Refine	×	81.9(0.3)	81.8(0.3)	64.9(0.8)	65.7(0.7)	63.1(0.3)	61.3(0.5)
	LLM-CARe(ours)	×	**82.6**(0.3)	**82.4**(0.4)	**85.3**(0.1)	**85.2**(0.1)	**76.9**(0.3)	**74.1**(0.5)

To evaluate the utility of synthetic EMRs, we assess their effectiveness in training models for downstream tasks. As shown in Table 2, **LLM-CARe achieves the best performance across all three tasks, without training or prompting on any real EMR text**. In contrast, most baselines rely on

表1：生成的电子病历（EMRs）在各原则上的质量评分（%），值越高表示性能越好。(*) 表示根据三个不同随机种子的运行计算的标准差。

Type	方法	依靠 EMR Text	章节级别	文档级		语料库级别	
			内容完整性	医疗正确性	上下文一致性	人口统计典型性	知识报道
Non-LLM	LSTM	✓	70.8(1.1)	65.0(0.4)	21.7(2.3)	93.3(0.6)	70.4(0.4)
	mtGAN	✓	55.8(2.9)	51.8(6.2)	21.4(4.2)	93.6(1.4)	76.3(3.7)
LLM-基于	医合 大型语言模型直连	✓	84.8(0.3)	95.3(0.8)	91.9(1.1)	84.1(0.9)	84.5(5.8)
		×	77.1(0.1)	90.7(0.2)	87.9(0.1)	77.7(0.1)	73.9(0.2)
	自我提升	×	78.3(0.2)	90.9(0.0)	88.5(0.1)	77.7(0.4)	78.0(2.4)
	LLM-CARe（我们的）	×	**91.2**(0.4)	**98.6**(0.0)	**93.8**(0.1)	**96.8**(1.4)	**94.1**(0.1)

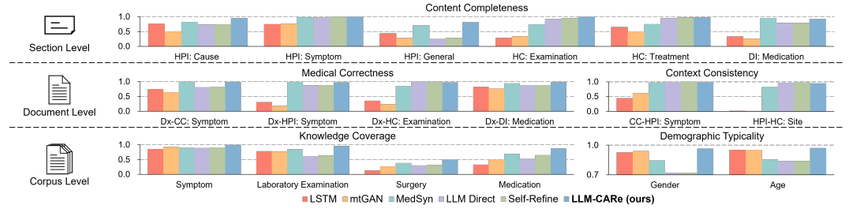


图5：跨三个层面的详细电子病历质量评估。缩写：CC-主诉，HPI-现病史，HC-住院经过，DI-出院指导，Dx-诊断。

图5展示了EMR质量在几个具有代表性且临床重要的标准上的细分情况；完整结果见附录H。我们的方法在大多数标准上达到最佳或具有竞争力的表现，显示了在多样化质量维度上的稳健性。值得注意的是，MedSyn在某些标准上甚至表现不如LLM直接方法，例如住院过程的完整性。这表明，MedSyn作为上下文示例使用的真实EMR可能存在遗漏，这些遗漏会传递到生成的记录中。这些发现进一步凸显了纯模仿方法的局限性，并强调在生成过程中明确建模和执行质量标准的重要性。

5.2 下游任务性能比较

表 2：三种下游任务的准确率（%），其中 micro 和 macro 为跨疾病的平均值。(*) 表示使用不同随机种子进行 3 次运行计算的标准差。

Type	方法	依靠 EMR Text	诊断预测		考试推荐		治疗推荐	
			微	宏	微	宏	微	宏
Non-LLM	LSTM	✓	74.0(2.0)	73.1(2.0)	75.7(0.3)	76.4(0.2)	56.7(0.6)	50.0(0.7)
	mtGAN	✓	81.9(2.2)	80.9(2.5)	72.4(1.5)	73.4(1.4)	58.6(2.8)	52.9(3.0)
LLM-基于	医合 大型语言模型直连	✓	81.7(0.0)	81.7(0.0)	82.9(0.1)	82.2(0.1)	74.5(0.1)	71.3(0.2)
		×	81.8(0.0)	81.8(0.2)	64.4(0.0)	65.4(0.0)	60.9(0.2)	59.0(0.3)
	自我提升	×	81.9(0.3)	81.8(0.3)	64.9(0.8)	65.7(0.7)	63.1(0.3)	61.3(0.5)
	LLM-CARe（我们的）	×	**82.6**(0.3)	**82.4**(0.4)	**85.3**(0.1)	**85.2**(0.1)	**76.9**(0.3)	**74.1**(0.5)

为了评估合成电子病历（EMR）的实用性，我们评估其在下游任务中训练模型的有效性。如表2所示，LLM-CARe在所有三项任务中均取得了最佳表现，而无需在任何真实电子病历文本上进行训练或提示。相比之下，大多数基线方法依赖于

real records for model training or in-context examples, which may raise privacy concerns (privacy evaluation is shown in Appendix I). While downstream tasks do not map directly to quality, the overall trend is consistent: methods with higher quality also tend to perform better on downstream tasks. For example, MedSyn ranks second to ours on both quality and tasks. Notably, baseline methods perform relatively well on diagnosis but show larger gaps on examination and treatment tasks. We attribute this to their higher coverage of symptom-related knowledge but limited representation of clinical concepts related to examinations, procedures, and medications—key to the latter two tasks. These results highlight the advantage of our structured framework in producing semantically rich and clinically useful synthetic records. Further experiments using an alternative backbone and an external dataset are provided in Appendix J, with subgroup analyses shown in Figure 13.

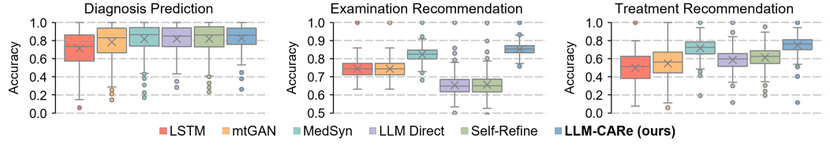


Figure 6: Accuracy distribution across diseases of different methods on three downstream tasks.

Figure 6 further illustrates the performance distribution across diseases. The box plots show that our method not only achieves higher average performance but also exhibits narrower variance across diseases. This consistency suggests that **our approach is broadly effective and robust across a wide range of clinical conditions**. In contrast, some baselines display wide performance fluctuations, indicating limited generalization to diverse disease types. These findings underscore the reliability of our method in real-world clinical settings, where robustness across varied diseases is critical.

5.3 Analysis of Performance Across Stages

Figure 7 shows how EMR quality and downstream task performance evolve through the three refinement stages. Quality dimensions improve most notably in their corresponding stages (e.g., completeness during the section stage). While some dimensions may show temporary fluctuations at other stages, the staged design—progressing from softer corpus-level constraints to stricter document-level checks—ensures that all dimensions ultimately exceed direct generation by a clear margin. For downstream tasks, examination and treatment recommendation benefit most from corpus-level refinement, since they rely on broad and diverse clinical concepts present in the training data. In contrast, diagnosis prediction depends more directly on complete histories and symptom–diagnosis alignment, thus improves primarily at the section and document stages.

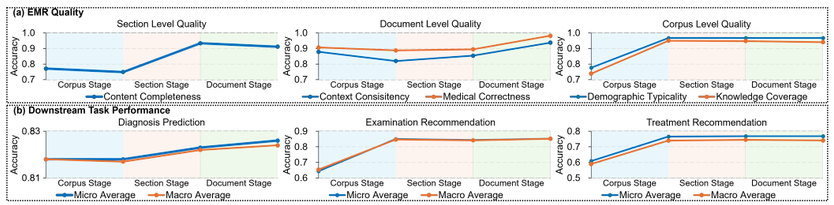
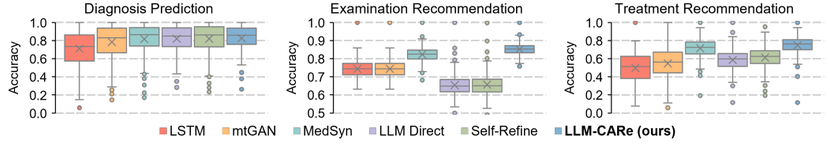


Figure 7: Trends on **(a)** EMR quality and **(b)** downstream task performance across stages.

5.4 Ablation Study of Multi-Agent Components

To assess the contribution of each agent in our framework, we conduct an ablation study by individually removing the Critic, Adviser, and Reviser agents. When the Critic is removed, the Adviser generates feedback for all quality criteria, regardless of whether they are already satisfied. Without

用于模型训练的真实记录或上下文示例，可能会引发隐私问题（隐私评估见附录I）。虽然下游任务并不直接对应质量，但总体趋势是一致的：质量较高的方法在下游任务上的表现也往往更好。例如，MedSyn在质量和任务上都仅次于我们的方法。值得注意的是，基线方法在诊断方面表现相对较好，但在检查和治疗任务上存在更大差距。我们将其归因于它们对症状相关知识的覆盖较高，但对与检查、操作和药物相关的临床概念表示有限——而这些概念对于后两个任务至关重要。这些结果凸显了我们结构化框架在生成语义丰富且临床有用的合成记录方面的优势。使用替代骨干模型和外部数据集进行的进一步实验见附录J，亚组分析见图13。



Fi图6：三种下游任务中不同方法在各疾病上的准确率分布。

图6进一步说明了不同疾病下的性能分布。箱线图显示，我们的方法不仅实现了更高的平均性能，而且在各疾病间表现出更小的方差。这种一致性表明我们的方法在广泛的临床条件下具有普遍的有效性和稳健性。相比之下，一些基线方法表现出较大的性能波动，表明其对多样疾病类型的泛化能力有限。这些发现强调了我们的方法在现实临床环境中的可靠性，在这里跨不同疾病的稳健性至关重要。

5.3 各阶段绩效分析

图7展示了EMR质量和下游任务性能在三个精炼阶段的变化情况。质量维度在其对应阶段的提升最为显著（例如，在章节阶段的完整性）。虽然某些维度在其他阶段可能出现暂时波动，但分阶段设计——从较宽松的语料库级约束逐步到更严格的文档级检查——确保所有维度最终都明显超过直接生成。对于下游任务，检查和治疗建议最受益于语料库级的精炼，因为它们依赖于训练数据中存在的广泛且多样的临床概念。相比之下，诊断预测更多依赖完整的病史和症状与诊断的匹配，因此主要在章节和文档阶段得到改进。

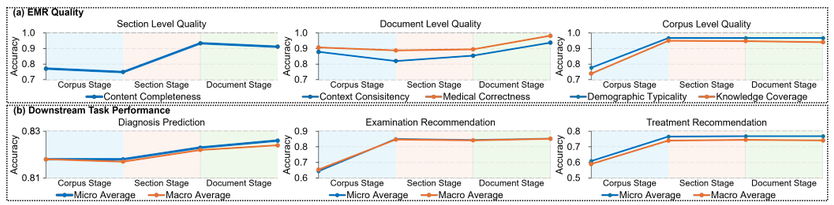


图7：（a）EMR质量和（b）下游任务性能在各阶段的趋势。

5.4 多智能体组件的消融研究

为了评估每个代理在我们框架中的贡献，我们通过单独移除评论者（Critic）、顾问（Adviser）和修订者（Reviser）代理来进行消融研究。当移除评论者时，顾问会针对所有质量标准生成反馈，而不管这些标准是否已经满足。没有

the Adviser, the Reviser receives only high-level information about unmet criteria, without actionable suggestions. When the Reviser is removed, the system cannot update existing drafts—instead, we prompt the Generator to regenerate EMRs using all quality criteria as input.

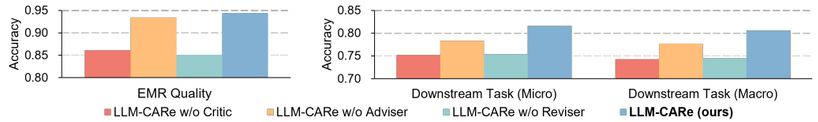


Figure 8: Impact of removing each agent on EMR quality and downstream task performance.

Figure 8 shows that removing any of the agents leads to a noticeable performance drop in both EMR quality and downstream tasks. The most significant declines occur when either the Critic or Reviser is ablated, highlighting two key insights: accurate assessment of the current draft is crucial for targeted refinement; and **large language models struggle to satisfy all quality criteria in a single generation step**, underscoring the need for cyclic refinement. Besides, removing the Adviser also results in a performance drop, suggesting that concrete, actionable feedback is more effective than abstract criterion-level input in guiding successful revisions.

5.5 Analysis of Robustness across LLM Backbone

Table 3 compares our framework with the LLM Direct baseline across three different LLM backbones: a general-purpose model (LLaMA 3.1) (Dubey et al., 2024), a medically pre-trained model (Meditron) (Chen et al., 2023), and a reasoning-oriented model (R1) (Guo et al., 2025). Without any prompt tuning or model-specific adaptation, our method consistently improves both EMR quality and downstream task performance across all backbones.

Table 3: EMR quality and downstream performance (%) across LLM backbones, averaged over principles and tasks.

Backbone	Generation Strategy	EMR Quality	Downstream Task Micro Average	Downstream Task Macro Average
Llama3.1 -8B-Instruct	LLM Direct	49.3	54.9	55.8
	LLM-CARe	**77.5**	**73.1**	**71.7**
Meditron3 -8B	LLM Direct	53.9	53.7	54.8
	LLM-CARe	**76.4**	**73.6**	**72.4**
R1-Distill -Llama-8B	LLM Direct	55.5	51.3	52.4
	LLM-CARe	**80.5**	**72.8**	**71.5**

Notably, **although Meditron is explicitly trained for medical domains, it still struggles to directly generate high-quality EMRs** and gains substantial improvements when integrated into our framework. Similarly, R1 does not significantly outperform the general model in direct generation, indicating that **internal reasoning alone is insufficient to meet the nuanced requirements of EMR**. These findings emphasize the necessity of principle-driven refinement that complements backbone capabilities and cannot be replaced by pretraining or reasoning alone.

5.6 Effect on Incorporating Real EMR Text

To further examine whether LLM-CARe benefits from access to real EMR text, we introduce a variant in which the initial draft generator is given a real EMR as a reference example. As illustrated in Figure 9, the performance of this variant remain highly similar to the original LLM-CARe. These findings show that the effectiveness of LLM-CARe arises from its structured multi-agent cyclic refinement, which leads to strong performance without relying on real EMR text. Additional comparisons with other variants are provided in Appendix K.

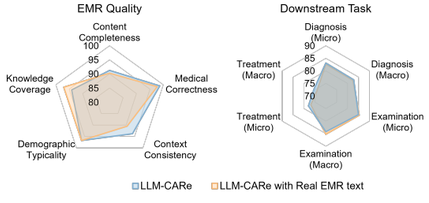


Figure 9: Comparison between LLM-CARe and a variant that incorporates real EMR text as an example during initial draft generation.

5.7 Clinician Evaluation

顾问、修改者仅接收关于未满足标准的高级信息，而没有可操作的建议。当修改者被移除时，系统无法更新现有的草稿——相反，我们会提示生成器使用所有质量标准作为输入来重新生成电子病历。

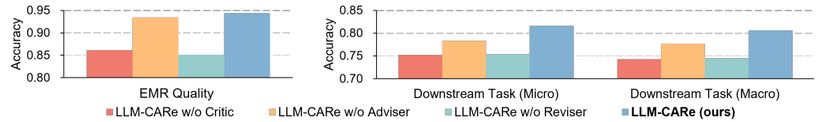


图 8：移除每个代理对电子病历质量及下游任务性能的影响 e.

图8显示，移除任何一个代理都会导致EMR质量和下游任务的性能显著下降。当消融Critic或Reviser时，下降最为明显，这突出了两个关键见解：准确评估当前草稿对于有针对性的改进至关重要；大语言模型在单次生成步骤中难以满足所有质量标准，强调了循环改进的必要性。此外，移除Adviser也会导致性能下降，这表明具体、可操作的反馈比抽象的标准层面输入在指导成功修订方面更有效。

5.5 跨大型语言模型骨干的鲁棒性分析

表3将我们的框架与LLM Direct基线在三种不同的LLM骨干模型上进行了比较：一个通用模型（LLaMA 3.1）（Dubey 等，2024）、一个医学预训练模型（Meditron）（Chen 等，2023）以及一个面向推理的模型（R1）（Guo 等，2025）。在没有任何提示调优或特定模型调整的情况下，我们的方法在所有骨干模型上都能持续提升EMR质量和下游任务性能。

表 3：不同大型语言模型骨干下 EMR 质量及下游性能（%），按原则和任务平均。

脊柱	一代策略	EMR 质量	下游任务 微平均	下游任务 宏平均
羊驼3.1 -8B-指导	大型语言模型直连	49.3	54.9	55.8
	LLM-CARe	**77.5**	**73.1**	**71.7**
Meditron3 -8B	大型语言模型直连	53.9	53.7	54.8
	LLM-CARe	**76.4**	**73.6**	**72.4**
R1-蒸馏 -羊驼-8B	大型语言模型直连	55.5	51.3	52.4
	LLM-CARe	**80.5**	**72.8**	**71.5**

值得注意的是，尽管Meditron明确接受过医疗领域的训练，它仍然在直接生成高质量电子病历（EMR）方面存在困难，但当集成到我们的框架中时会获得显著改进。同样，R1在直接生成方面并未显著优于通用模型，这表明仅靠内部推理不足以满足电子病历的细致要求。这些发现强调了基于原则的精炼的必要性，这种精炼能够补充主干能力，而不能仅靠预训练或推理来替代。

5.6 纳入真实电子病历文本的影响

为了进一步研究 LLM-CARe 是否从访问真实 EMR 文本中受益，我们引入了一个变体，其中初始草稿生成器以真实 EMR 作为参考示例。如图 9 所示，该变体的性能与原始 LLM-CARe 高度相似。这些发现表明，LLM-CARe 的有效性来自其结构化多代理循环优化，从而无需依赖真实 EMR 文本也能实现强大的性能。附录 K 提供了与其他变体的额外比较。

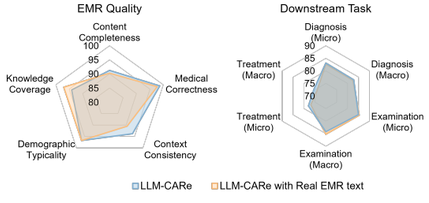


图9：LLM-CARe 与在初稿生成过程中将真实电子病历文本作为示例的变体之间的比较。

5.7 临床医生评估

To validate the reliability of using LLM as a judge, we conducted a human evaluation study. A total of 200 synthetic EMRs were sampled (40 from each of five methods) and independently assessed by four licensed clinicians, who rated completeness, consistency, and correctness for each record. As shown in Table 4, the agreement between clinicians and LLM is consistently high (Cohen's Kappa = 0.833 overall, where values exceeding 0.8 indicate near-perfect agreement), with tight confidence intervals. Inter-clinician agreement is also strong (Fleiss's Kappa = 0.950 overall), confirming that the evaluation criteria are well-defined and consistently interpretable by human experts. Together, these results demonstrate that the **LLM-based evaluation closely aligns with human judgment, supporting its validity as an efficient proxy for large-scale quality assessment**. To ensure the stability of human evaluation, we gradually expanded the annotated subset from 100 to 200 EMRs and observed consistently stable agreement levels. Detailed results are reported in Appendix L.

Table 4: Agreement between human clinicians and LLM-based evaluation on EMR quality.

Quality Level	Clinician-LLM Agreement		Inter-Clinician Agreement	
	Cohen's Kappa	Confidence Interval (95%)	Fleiss's Kappa	Confidence Interval (95%)
Section	0.866	[0.834, 0.896]	0.975	[0.959, 0.988]
Document	0.797	[0.764, 0.829]	0.932	[0.912, 0.950]
Overall	0.833	[0.809, 0.855]	0.950	[0.937, 0.962]

5.8 Case Study

Table 5 presents examples of quality issues that commonly arise when generation methods lack explicit adherence to quality standards. These cases reveal that without structured quality control, generated EMRs often exhibit missing details, medical inaccuracies, or inconsistencies.

In contrast, Figure 10 demonstrates how LLM-CARe progressively improves draft quality through refinement on different levels. This underscores the importance of stage-wise cyclic refinement in producing high-quality EMRs.

Table 5: Examples of quality issues in synthetic EMRs. Abbreviations: CC-Chief Complaint, HPI-History of Present Illness, HC-Hospital Course, Dx-Diagnosis.

Method	Example	Problem
LSTM	**Dx**: Uterine leiomyomas **Gender**: Male	Males do not have a uterus.
mtGAN	**HC**: Discharged after feeling stable.	No treatments are mentioned in HC.
MedSyn	**CC**: Diarrhea for 2 days. **HPI**: ... no diarrhea ...	CC mentions diarrhea, but HPI denies it.

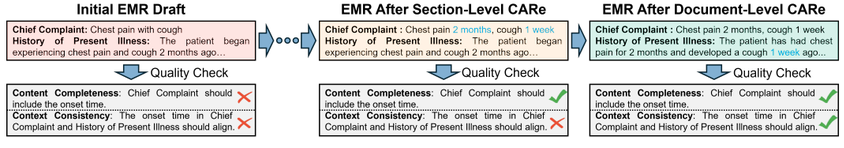


Figure 10: Illustration of quality improvements through LLM-CARe. Revisions are marked in blue.

6 Conclusion

In this work, we tackle the limitations of existing EMR synthesis methods which mimic real records without considering quality requirements. To overcome these, we propose **LLM-CARe**, a stage-wise cyclic refinement framework driven by the collaboration of **Critic**, **Adviser**, and **Reviser** agents. Instead of single-pass generation, LLM-CARe progressively enhances drafts through three dedicated stages: aligning corpus-level distributions, ensuring section-level completeness, and enforcing document-level consistency and correctness. Experiments on a large real-world dataset demonstrate that LLM-CARe substantially improves the quality of EMRs across all granularities. Moreover, models trained on the refined synthetic corpus achieve superior performance on various downstream tasks, highlighting the practical value of our approach. These results show the effectiveness of LLM-CARe in generating synthetic EMRs that are both high-quality and clinically meaningful, offering a reliable and privacy-preserving foundation for healthcare AI development.

为了验证使用大型语言模型（LLM）作为评判者的可靠性，我们进行了人工评估研究。总共抽取了200份合成电子病历（每种方法抽取40份，共五种方法），由四位有执照的临床医师独立评估，每位记录的完整性、一致性和正确性如表4所示，评估结果的一致性临床医生与大型语言模型（LLM）之间的一致性始终很高（Cohen's Kappa = 0总体为0.833，其中超过0.8的值表示近乎完美的一致性），且置信区间较窄。临床医生之间的一致性也很强（Fleiss's Kappa = 0总体为0.950），这证明了评估标准定义明确，并且人类专家能够始终如一地理解。综合来看，这些结果表明基于LLM的评估与人类判断高度一致，支持其作为大规模质量评估的高效替代方法的有效性。为了确保人工评估的稳定性，我们将标注子集逐步从100份EMR扩展到200份，并观察到一致性水平始终稳定。详细结果在附录L中报告。

表 4：人类临床医生与基于大语言模型的电子病历质量评估之间的一致性。

质量水平	临床医生-大语言模型协议		临床医生间协议	
	科恩的卡帕	自信区间（95%）	弗莱斯的卡帕	自信区间（95%）
部分	0.866	[0.834, 0.896]	0.975	[0.959, 0.988]
文件	0.797	[0.764, 0.829]	0.932	[0.912, 0.950]
总体上	0.833	[0.809, 0.855]	0.950	[0.937, 0.962]

5.8 案例研究

表5列出了当生成方法缺乏对质量标准的明确遵循时常出现的质量问题示例。这些案例显示，如果没有结构化的质量控制，生成的电子病历通常会出现细节缺失、医学不准确或不一致的情况。

相比之下，图10展示了LLM-CARe如何通过在不同层面上的优化逐步提高草稿质量。这强调了其重要性在生成高质量电子病历时采用分阶段循环精炼。

表 5：合成电子病历中的质量问题示例。缩写：CC-主诉，HPI-现病史，HC-住院过程，Dx-诊断。

方法	例子	问题
LSTM	诊断：子宫平滑肌瘤 **性别：男**	男性不有子宫。
mtGAN	出院后感觉稳定。	没有治疗方法在HC中提到。
医合	主诉：腹泻持续2天。 病史现病史：……没有腹泻…	主诉提到腹泻，但病史否认这一点。

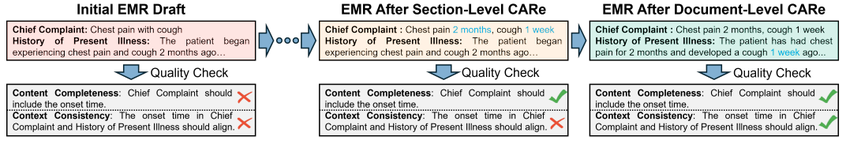


图10：通过LLM-CARe实现质量改进的示意图。修订部分以蓝色标出。

6 结论

在本研究中，我们处理了现有电子病历（EMR）合成方法的局限性，这些方法在模拟真实病历时未考虑质量要求。为克服这些问题，我们提出了LLM-CARe，一种由评论者（Critic）、顾问（Adviser）和修订者（Reviser）代理协作驱动的分阶段循环优化框架。LLM-CARe不是一次性生成，而是通过三个专门阶段逐步提升草稿：对齐语料库级分布、确保章节级完整性以及强化文档级一致性和正确性。在大型真实世界数据集上的实验表明，LLM-CARe显著提高了各个粒度层面的电子病历质量。此外，在经过优化的合成语料上训练的模型在各种下游任务中取得了更优异的表现，凸显了我们方法的实际价值。这些结果表明，LLM-CARe在生成既高质量又具有临床意义的合成电子病历方面是有效的，提供了一种可靠且隐私

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ETHICS STATEMENT

This work adheres to the ICLR Code of Ethics. This work focuses on improving the quality of synthetic EMRs guided by clinical quality principles. The core methodology does not involve training on actual EMRs. During evaluation, a limited set of test cases was accessed within a secure, institutional data environment. These records had been fully de-identified by the hosting healthcare organization and remained within its controlled data management platform. The study did not entail any active data collection from patients or clinicians. All data usage adhered to institutional policies and was conducted under the oversight of the relevant data governance framework.

REPRODUCIBILITY STATEMENT

The collection and preprocessing of the EMR dataset are described in Section 4 and Appendix B. Experimental settings, model configurations, and evaluation protocols are detailed in Section 4 and Appendix E. Our code is available at `https://github.com/THUMLP/LLM-CARe`.

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伦理声明

本工作遵守 ICLR 伦理规范。本工作重点在于通过临床质量原则指导，提高合成电子病历（EMR）的质量。核心方法不涉及基于实际电子病历的训练。在评估过程中，仅访问了有限的一组测试案例，这些案例位于安全的机构数据环境中。这些记录已由托管的医疗机构完全去标识化，并保持在其受控的数据管理平台内。该研究未涉及任何来自患者或临床医生的主动数据收集。所有数据使用均遵守机构政策，并在相关数据治理框架的监督下进行。

可重复性声明

电子病历（EMR）数据集的收集与预处理在第4节和附录B中进行了描述。实验设置、模型配置和评估协议在第4节和附录E中详细说明。我们的代码可在 https://github.com/THUMLP/LLM-CARe 获取。

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A Details of EMR Quality Principles

In this section, we provide a detailed description of each criterion corresponding to the EMR quality principles, along with representative examples.

A.1 Content Completeness

Table 6 lists the criteria used to assess content completeness, which evaluate whether each field contains all essential information. The criteria follow standard clinical documentation conventions and use ordinary clinical terms (e.g., "major symptoms" refers to the primary complaints driving the visit, as opposed to secondary associated manifestations).

Table 6: Description of quality criteria for content completeness, along with representative examples. Abbreviations: CC-Chief Complaint, HPI-History of Present Illness, HC-Hospital Course, DI-Discharge Instructions.

Criterion	Abbreviation	Positive Example	Negative Example
Chief complaint states reason for admission	CC Reason	• CC: Cough for 1 day • CC: Thyroid nodule noted for 2 months	• CC: Admitted on 2025/05/16
Chief complaint includes onset time	CC Onset	• CC: Fever for 6 days • CC: Chest pain for 1 year, worsened over past month	• CC: Dizziness accompanied by nausea • CC: Poor recent glycemic control
History of present illness describes acuity of onset	HPI Acuity	• HPI: Sudden-onset headache 1 week ago • HPI: Gradual onset of unsteady gait for 4 months	• HPI: Experienced headache over a month ago • HPI: Developed gait instability recently
History of present illness mentions possible causes	HPI Cause	• HPI: Abdominal pain after alcohol intake 1 day ago • HPI: Dizziness for 2 weeks without obvious cause	• HPI: Sudden right eye vision loss one week ago • HPI: Cough onset 3 days ago
History of present illness lists major symptoms and onset time	HPI Symptom	• HPI: Vomited 4–5 times over the past half day • HPI: Poor appetite and fatigue over past 2 weeks	• HPI: Experienced dizziness for days
History of present illness includes all general conditions	HPI General	• HPI: Normal mental status, sleep, appetite, bowel and bladder function; no significant weight change	• HPI: Appetite decreased
Hospital course includes auxiliary examinations or laboratory examinations	HC Examination	• HC: Chest CT revealed a pulmonary mass lesion • HC: Admission labs showed CRP: 12.3 mg/L	• HC: Patient underwent further examinations after admission
Hospital course includes treatment interventions	HC Treatment	• HC: Appendectomy under general anesthesia • HC: Aspirin given for antiplatelet therapy	• HC: Given pharmacological therapy
Discharge instruction includes medication dosage and usage	DI Medication	• DI: Atorvastatin 1 tablet nightly • DI: Amoxicillin 1g twice daily	• DI: Take antibiotics regularly

A.2 Medical Correctness

Table 7 outlines the criteria for medical correctness, which assess whether the clinical content aligns with the patient's diagnosis.

电子病历质量原则的详细信息

在本节中，我们提供了每个与电子病历质量原则相对应的标准的详细说明，以及具有代表性的示例。

A.1 内容完整性

表6列出了用于评估内容完整性的标准，这些标准用于评估每个字段是否包含所有必要信息。该标准遵循标准临床文档规范，并使用普通临床术语（例如，“主要症状”指引起就诊的主要投诉，而不是次要相关表现）。

表6：内容完整性质量标准的描述及代表性示例。缩写：CC-主诉，HPI-现病史，HC-住院经过，DI-出院指导。

标准	缩写	积极的例子	负面例子
主诉 说明理由 入场	CC 原因	• CC：咳嗽1天 • CC：甲状腺结节已记录两个月	• CC：入院时间 2025/05/16
主诉 包括发作时间	CC 开始	• 主诉：发热6天 • 主诉：胸痛已有1年在过去一个月中恶化	• 主诉：头晕伴有恶心 • CC：近期状况不佳血糖控制
现在史 疾病描述 发作急性	HPI 敏锐	• 病史：突然发作一周前头痛 • 病史：逐渐出现步态不稳已持续4个月	• 病史：有经验一个月前的头痛 • 现病史：出现步态异常最近的不稳定
现在史 疾病提及 可能的原因	HPI 原因	• 病史：腹痛之后1天前饮酒 • 现病史：头晕持续2周没有明显原因	• 现病史：右眼突然一周前视力丧失 • 现病史：咳嗽出现 3 几天前
现在史 疾病列表 主要症状 以及发作时间	HPI 症状	• 现病史：呕吐4-5次在过去半天里 • 现病史：食欲差且过去两周的疲劳	• 病史：有经验头晕好几天
现在史 疾病包括 所有一般条款	HPI 一般	• 现病史：精神状态正常状态、睡眠、食欲、排便以及膀胱功能；无显著体重变化	• 主诉：食欲减少
住院经过包括 辅助检查 或实验室检查	HC 考试	• HC：胸部CT显示肺部肿块病变 • HC：入院实验室检查显示 C 反应蛋白：12.3 毫克/升	• HC：患者接受了进一步检查入院后
住院过程 包括治疗 干预	HC 治疗	• HC：阑尾切除术在全身麻醉下 • HC：给予阿司匹林用于抗血小板治疗	• HC：给定药物治疗
出院指导 包括药物 剂量和用法	DI 药物	• DI：阿托伐他汀每晚 1 片 • DI：阿莫西林每天两次，每次1克	• DI：服用抗生素定期地

A.2 医学正确性

表7概述了医学正确性的标准，这些标准用于评估临床内容是否与患者的诊断一致。

Table 7: Description of quality criteria for medical correctness, along with representative examples. Abbreviations: CC-Chief Complaint, HPI-History of Present Illness, HC-Hospital Course, DI-Discharge Instructions, Dx-Diagnosis, PD-Patient Demographics.

Criterion	Abbreviation	Positive Example	Negative Example
Diagnosis matches the patient's gender	Dx-PD Gender	• Dx: Pregnancy Gender: Female	• Dx: Pregnancy Gender: Male
Symptoms in chief complaint align with diagnosis	Dx-CC Symptom	• Dx: Pneumonia CC: Cough for 1 day	• Dx: Pneumonia CC: Knee pain for 3 days
Symptoms in history of present illness align with diagnosis	Dx-HPI Symptom	• Dx: Cerebral infarction HPI: Sudden slurred speech 1 day ago	• Dx: Acute appendicitis HPI: Sudden blurred vision 2 weeks ago
Examinations in hospital course align with diagnosis	Dx-HC Examination	• Dx: Pneumonia HC: Chest CT indicated pneumonia	• Dx: Cerebral infarction HC: Abdominal ultrasound showed appendiceal thickening
Medications in discharge instructions align with diagnosis	Dx-DI Medication	• Dx: Type 2 diabetes DI: Metformin (0.5g), one tablet twice daily	• Dx: Pneumonia DI: Insulin injection before meals

A.3 Context Consistency

Table 8 presents the criteria for context consistency, which evaluate whether information across different EMR sections is logically coherent.

Table 8: Description of quality criteria for context consistency, along with representative examples. Abbreviations: CC-Chief Complaint, HPI-History of Present Illness, HC-Hospital Course.

Criterion	Abbreviation	Positive Example	Negative Example
Symptoms in chief complaint are consistent with those in history of present illness	CC-HPI Symptom	• CC: Cough for 1 day HPI: ... wich cough ...	• CC: Cough for 1 day HPI: ... wichout cough ...
Onset time in chief complaint is consistent with that in history of present illness	CC-HPI Onset	• CC: Chest pain for 1 month HPI: Chest pain over past 1 month	• CC: Chest pain for 1 month HPI: Chest pain over past 2 months
Affected site in history of present illness is consistent with the site of examination or treatment in hospital course	HPI-HC Site	• HPI: Left leg pain after a fall HC: X-ray showed a fracture of the left leg.	• HPI: Left leg pain after a fall HC: X-ray showed a fracture of the right leg.

A.4 Demographic Typicality

For demographic typicality, we focus on two key patient attributes: gender and age. We evaluate whether the distributions of these attributes in the synthetic EMRs align with the target distributions. Figure 11 illustrates representative examples of gender and age distributions that are aligned with and deviate from the target distribution.

表7：医疗正确性质量标准的描述及代表性示例。缩写：CC-主诉，HPI-现病史，HC-住院过程，DI-出院指导，Dx-诊断，PD-患者人口统计信息。

标准	缩写	积极的例子	负面例子
诊断匹配患者的性别	Dx-PD 性别	• Dx：妊娠 性别：女	• Dx：妊娠 性别：男
症状在主诉与诊断一致	Dx-CC 症状	• 诊断：肺炎 主诉：咳嗽1天	• 诊断：肺炎 主诉：膝盖疼痛3天
病史中的症状现病史与诊断一致	Dx-HPI 症状	• Dx：大脑梗死 病史现病史：突然口齿不清 演讲 1 天前	• Dx：急性阑尾炎 病史：突然视力模糊 两周前的愿景
考试在住院过程与诊断一致	Dx-HC 考试	• 诊断：肺炎 HC：胸部CT显示肺炎	• Dx：大脑梗死 HC：腹部超声显示阑尾增厚
药物在出院指导与诊断一致	Dx-DI 药物	• 诊断：2型糖尿病 DI：二甲双胍（0.5克），每天两次，每次一片	• 诊断：肺炎 DI：胰岛素注射饭前

A.3 上下文一致性

表8展示了上下文一致性的标准，用于评估不同电子病历（EMR）部分的信息是否在逻辑上连贯。

表8：上下文一致性质量标准的描述及代表性示例。缩写：CC-主诉，HPI-现病史，HC-住院经过。

标准	缩写	积极的例子	负面例子
主要症状投诉是一致的与历史上的那些人现病史	主诉与现病史 症状	• CC：咳嗽1天 病史现病史：……咳嗽……	• CC：咳嗽1天 现病史：……没有咳嗽……
主要症状的起始时间抱怨是一贯的与此相关的历史现病史	主诉与现病史 开始	• 主诉：胸痛持续1个月 现病史：胸痛持续过去1个月	• 主诉：胸痛持续1个月 病史现病史：胸痛超过过去2个月
既往病变部位现病史一致在检查部位或住院期间的治疗	HPI-HC Site	• 现病史：左腿疼痛摔倒之后 HC：X光显示左腿骨折。	• 现病史：左腿疼痛摔倒之后 HC：X光显示右腿骨折。

A.4 人口统计典型性

为了人口统计学的典型性，我们关注两个关键的患者属性：性别和年龄。我们评估这些属性在合成电子病历中的分布是否与目标分布一致。图11展示了与目标分布一致和偏离目标分布的性别和年龄分布的代表性示例。

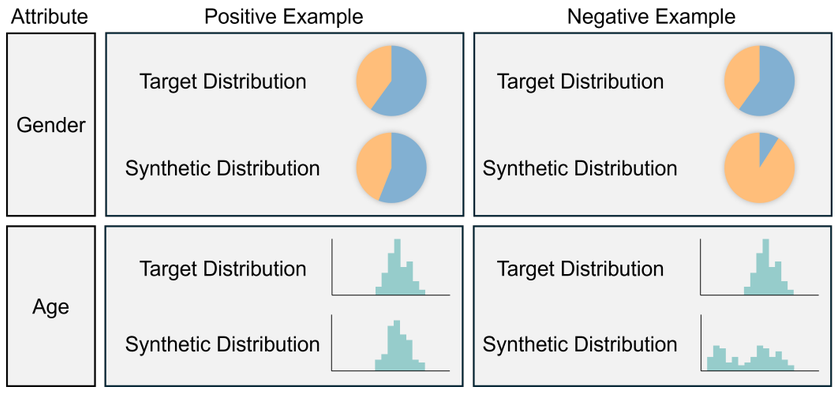


Figure 11: Examples of synthetic data distributions that are either consistent or inconsistent with the target distribution.

A.5 Knowledge Coverage

For knowledge coverage, we focus on five key categories of clinical knowledge: symptoms, auxiliary examinations, laboratory examinations, surgeries, and medications. Table 9 lists representative entities from each category.

Table 9: Knowledge categories and representative entities.

Category	Example
Symptom	Cough, Fever, Headache, Nausea, ...
Auxiliary Examination	Chest CT, Brain MRI, ECG, Abdominal Ultrasound, ...
Laboratory Examination	Complete Blood Count, Liver Function Test, C-Reactive Protein, ...
Surgery	Appendectomy, Tonsillectomy, Cataract Surgery, Cholecystectomy, ...
Medication	Aspirin, Penicillin, Metformin, Atorvastatin, ...

B Details of Dataset

B.1 Dataset Construction

We conduct our experiments on a large-scale real-world EMR dataset containing 1.82 million de-identified medical records collected from hospitals. Personally identifiable information (e.g., patient and clinician names, phone numbers, locations) had already been removed by the data provider, ensuring compliance with privacy standards. All experiments were conducted on hospital-controlled infrastructure to ensure data security and prevent risk of privacy leakage.

To ensure data quality, we first remove records that are missing critical information, such as patient age or gender, primary diagnosis, or any of the four target fields: chief complaint, history of present illness, hospital course, and discharge instruction. After this filtering step, 905k records remain.

We then apply a length-based filtering criterion to further improve data quality. During inspection, we found that overly short entries often contain placeholders or incomplete content, while excessively long entries are more likely to include unintelligible text. Therefore, we retain only records where the chief complaint is under 20 words, and each of the other fields falls within the 10 to 1,000 word range. This step yields a subset of 710k high-quality records.

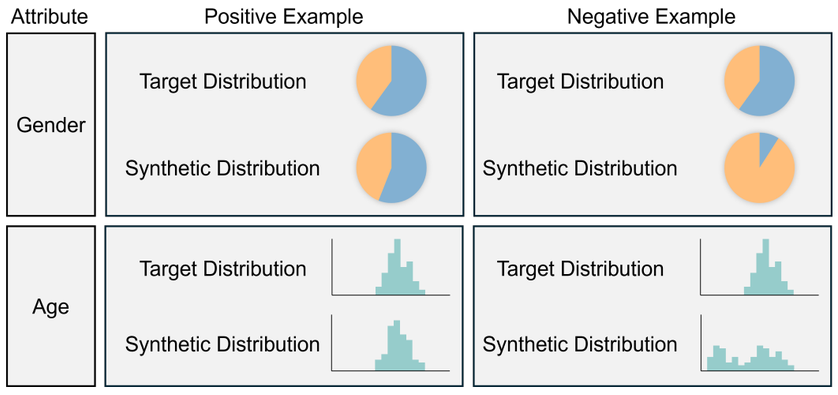


图11：与目标分布一致或不一致的合成数据分布示例。

A.5 知识覆盖

在知识覆盖方面，我们关注临床知识的五个关键类别：症状、辅助检查、实验室检查、手术和药物。表9列出了每个类别的代表性实体。

表9：知识类别及代表实体。

类别	例子
症状	咳嗽，发烧，头痛，恶心，…
辅助检查	胸部CT，脑部MRI，心电图，腹部超声，…
实验室检查	全血细胞计数，肝功能测试，C-反应蛋白，…
手术	阑尾切除术、扁桃体切除术、白内障手术、胆囊切除术、…
药物	阿司匹林，青霉素，二甲双胍，阿托伐他汀，…

B 数据集详情

B.1 数据集构建

我们在一个大规模的真实世界电子病历（EMR）数据集上进行实验，该数据集包含从医院收集的182万条去标识化的病历。数据提供方已经删除了个人可识别信息（例如患者和医生姓名、电话号码、地点），以确保符合隐私标准。所有实验均在医院控制的基础设施上进行，以确保数据安全并防止隐私泄露的风险。

为了确保数据质量，我们首先删除缺少关键信息的记录，例如患者年龄或性别、主要诊断，或四个目标字段中的任何一个：主诉、现病史、住院经过和出院指导。经过此过滤步骤后，剩余 905k 条记录。

然后，我们应用基于长度的过滤标准以进一步提高数据质量。在检查过程中，我们发现过短的条目通常包含占位符或不完整内容，而过长的条目更可能包含难以理解的文本。因此，我们仅保留主诉少于20个单词的记录，并且其他各字段在10到1000个单词范围内的记录。此步骤得到了一组71万条高质量记录的子集。

Lastly, to ensure the reliability and stability of downstream evaluation, we retain only common diseases with sufficient data volume. Specifically, we exclude any diagnosis category with fewer than 500 records. This ensures that each disease has at least 100 samples in the test set after an 80/20 train-test split. Diseases with very few examples can lead to high-variance estimates and lack statistical significance in evaluation. After this final filtering step, we obtain 192k EMRs spanning 302 distinct disease categories.

B.2 Construction of Downstream Tasks

For all downstream tasks, the correct options are extracted directly from the EMR. For the diagnosis prediction task, incorrect options are randomly sampled from other diagnoses in the dataset. For the test and treatment prediction tasks, incorrect options are selected to be incompatible with the gold diagnosis: we first exclude all tests or treatments that appear in EMRs with the same diagnosis, and then randomly sample from the remaining pool. Using this approach, we construct training examples from synthetic EMRs, and evaluate model performance on questions built from real EMRs in the held-out test set. The diagnosis prediction task contains exactly one question per EMR, yielding 38k samples in the test set. For examination and treatment recommendation, each extracted examination or treatment entity forms one question, resulting in 346k and 110k test questions respectively.

C Baseline Implementation Details

For the non-LLM baselines (LSTM and mtGAN), we trained both models from scratch on the training set. Each model receives a one-hot disease vector as input and is optimized to generate the corresponding real EMR, treated as the supervision target.

For the LLM-based baselines (MedSyn, LLM Direct, and Self-Refine), we used the same backbone model as in LLM-CARe (Qwen2.5-7B-Instruct) to ensure a fair comparison. **MedSyn**: For each generation, we randomly sample one real EMR of the target disease from the training set as an in-context example, and randomly sample five symptoms associated with the disease as additional cues. The model then generates an EMR in a single pass conditioned on these inputs. **LLM Direct**: We generate EMRs using the same initial-draft prompt template as in LLM-CARe, without applying any refinement or evaluation steps. **Self-Refine**: We aggregate all section-level and document-level criteria into a single combined instruction. The model first identifies issues based on this unified prompt and then produces a revised EMR accordingly, without multi-level decomposition.

D Evaluation Details

For the LLM-based evaluation of EMR quality, we prompt the model to assess each generated EMR against the predefined criteria for medical correctness, content completeness, and context consistency. Each criterion is formulated as a binary classification task—whether a given EMR satisfies the criterion or not. The model outputs a yes/no response for each criterion per EMR, and we compute the final score by averaging over all EMRs.

For demographic typicality, we compare the distribution of demographic attributes in synthetic EMRs to those in the real dataset. For gender, we use the total variation distance (TVD) between the two distributions. For age, which is a continuous variable, we compute the Wasserstein distance. Since lower distance values indicate higher similarity, we transform the scores by computing $1 - \text{TVD}$ and $1 - \text{Wasserstein}$, respectively, so that higher values consistently reflect better quality across all metrics.

For knowledge coverage, we first extract medical entities associated with each diagnosis from the real EMRs. We then measure the proportion of these entities that appear in synthetic EMRs with the same diagnosis. To avoid the complexity and potential noise of semantic matching, we use exact string-level matching to compare entity presence.

For the downstream task evaluation, we report both macro and micro accuracy. Macro accuracy averages the model performance across all diagnoses by first computing the accuracy within each disease category, then averaging across categories. Micro accuracy, in contrast, computes the overall

最后，为了确保下游评估的可靠性和稳定性，我们仅保留数据量充足的常见疾病。具体来说，我们排除任何记录数量少于500的诊断类别。这样可以确保在80/20训练-测试集划分后，每种疾病在测试集中至少有100个样本。样本非常少的疾病可能导致高方差估计，并且在评估中缺乏统计显著性。经过这一最终筛选步骤后，我们获得了覆盖302个不同疾病类别的19.2万条电子病历。

B.2 下游任务的构建

对于所有下游任务，正确选项都是直接从电子病历（EMR）中提取的。对于诊断预测任务，错误选项是从数据集中的其他诊断中随机抽取的。对于检查和治疗预测任务，错误选项的选择是为了与黄金诊断不兼容：我们首先排除出现在相同诊断的电子病历中的所有检查或治疗，然后从剩余的池中随机抽取。通过这种方法，我们从合成电子病历构建训练样本，并在保留的测试集中的真实电子病历生成的问题上评估模型性能。诊断预测任务每份电子病历恰好包含一个问题，测试集中共有38k个样本。对于检查和治疗推荐，每个提取的检查或治疗实体形成一个问题，分别产生346k和110k个测试问题。

C 基线实现细节

对于非LLM基线（LSTM和mtGAN），我们从零开始在训练集上训练这两个模型。每个模型都接收一个独热疾病向量作为输入，并被优化以生成相应的真实电子病历，作为监督目标。

对于基于LLM的基线方法（MedSyn、LLM Direct和Self-Refine），我们使用了与LLM-CARe相同的骨干模型（Qwen2.5-7B-Instruct）以确保公平比较。MedSyn：在每次生成时，我们从训练集中随机抽取一个目标疾病的真实电子病历作为上下文示例，并随机抽取该疾病相关的五个症状作为附加提示。模型随后根据这些输入一次性生成电子病历。LLM Direct：我们使用与LLM-CARe相同的初稿提示模板生成电子病历，不进行任何改进或评估步骤。Self-Refine：我们将所有章节级和文档级的标准汇总为单一的综合指令。模型首先根据这一统一提示识别问题，然后生成相应的修订版电子病历，无需多层级分解。

D 评估详情

对于基于大语言模型（LLM）的电子病历（EMR）质量评估，我们提示模型根据预定义的医疗正确性、内容完整性和上下文一致性标准评估每份生成的电子病历。每个标准被制定为一个二元分类任务——判断给定的电子病历是否满足该标准。模型为每份电子病历的每个标准输出是/否回答，我们通过对所有电子病历的得分取平均值来计算最终得分。

对于人口统计学典型性，我们将合成电子病历中的人口统计属性分布与真实数据集进行比较。对于性别，我们使用两种分布之间的总变差距离（TVD）。对于年龄这一连续变量，我们计算瓦瑟斯坦距离。由于较低的距离值表示更高的相似性，我们通过分别计算 $1 - \text{TVD}$ 和 $1 - \text{Wasserstein}$ 对分数进行转换，从而使所有指标中更高的值始终反映更高的质量。

为了知识覆盖，我们首先从真实的电子病历中提取与每个诊断相关的医学实体。然后，我们测量这些实体在具有相同诊断的合成电子病历中出现的比例。为了避免语义匹配的复杂性和潜在噪声，我们使用精确的字符串级匹配来比较实体的存在情况。

对于下游任务评估，我们报告宏观和微观准确率。宏观准确率通过首先计算每个疾病类别内的准确率，然后在各类别之间求平均，来对模型在所有诊断中的表现取平均值。相比之下，微观准确率计算整体准确率

accuracy across all samples regardless of diagnosis. This dual evaluation provides a comprehensive view of model generalizability across frequent and less frequent disease types.

E EXPERIMENTAL DETAILS

E.1 PROMPTS USED FOR EACH AGENT

In this section, we list the prompts used for each agent in our LLM-CARe framework.

Generator

> Please generate an electronic medical record according to the following requirements:
> 1. The patient's primary diagnosis is: [diag].
> 2. Include only the following sections: 'Gender', 'Age', 'Primary Diagnosis', 'Chief Complaint', 'History of Present Illness', 'Hospital Course', and 'Discharge Instructions'.
> 3. Section-specific instructions:
> - The Chief Complaint should briefly describe the reason for admission.
> - The History of Present Illness should describe the onset and development of the condition in detail.
> - The Hospital Course should mention the examinations and treatments the patient received.
> - The Discharge Instructions should specify post-discharge recommendations, such as prescribed medications.
> 4. Output the result in JSON format with the structure: "Section Name": "Section Content", where each section content is a single string.

Section-Level Critic

> Below is the '[section_name]' section from an electronic medical record:
> [section]
> Please determine whether the above '[section_name]' meets the following requirement: [requirement].
> Respond in the following JSON format:
> {"Meets Requirement": true/false}

Section-Level Adviser

> Below is the '[section_name]' section from an electronic medical record:
> [section]
> This section does not meet the following requirement: [requirement]. Please provide a specific revision suggestion based on the section content, explaining how it should be modified to meet the requirement.
> Respond in the following JSON format:
> {"Revision Suggestion": "specific suggestion"}

Section-Level Reviser

> Below is the '[section_name]' section from an electronic medical record with an issue:
> [section]
> The '[section_name]' section misses essential content. Please revise the record based on the following suggestion: [feedback]
> Return the result in JSON format using the pattern "Section Name": "Section Content", and include only the "[section_name]" section. The content of the section should be a single string.

无论诊断如何，所有样本的准确性。这种双重评估提供了对模型在常见和不常见疾病类型上的泛化能力的全面视角。

E 实验细节

E.1 每个代理使用的提示

在本节中，我们列出了在我们的 LLM-CARe 框架中为每个代理使用的提示。

生成器

> 请根据以下要求生成一份电子病历：1. 患者的主要诊断为： [diag]。2. 仅包括以下部分：‘性别’、‘年龄’、‘主要诊断’、‘主诉’、‘现病史’、‘住院经过’和‘出院医嘱’。3. 各部分具体要求： - 主诉应简要描述入院原因。 - 现病史应详细描述病情的起病和发展情况。 - 住院经过应提及患者接受的检查和治疗。 - 出院医嘱应明确出院后的建议，例如所开处方药物。4. 以JSON格式输出结果，结构为：“部分名称”：“部分内容”，其中每个部分内容为单个字符串。

章节级评论者

> 以下是电子病历中的‘[section name]’部分：[section]请确定上述‘[section name]’是否符合以下要求： [requirement]。请按以下 JSON 格式回复：{“符合要求”： true/false}。

科目级顾问

> 以下是电子病历中的’[部分名称]’部分：[部分]该部分不符合以下要求： [要求]。请根据该部分内容提供具体的修改建议，说明应如何修改以满足要求。请以以下JSON格式回复：{”修改建议”：”具体建议”}

章节级修订者

> 下面是电子病历中‘[section name]’部分的一个问题：
>
> [部分]
> “[部分名称]”部分缺少必要内容。请根据以下建议修改记录： [反馈]
>
> 以 JSON 格式返回结果，使用模式“Section Name”：“Section Content”，并仅包含“[section name]”部分。- 该部分的内容应为单个字符串。

Document-Level Critic

> Below is an electronic medical record consisting of multiple sections:
>
> [record]
>
> Please evaluate whether the record satisfies the following requirement: [requirement]. Identify any conflicts or implausible statements across sections.
>
> Respond in the following JSON format:
>
> {"Meets Requirement": true/false}

Document-Level Adviser

> The following record has issues violating the requirement: [requirement]:
>
> [record]
>
> Please provide targeted suggestions to resolve the problem. Prioritize changes to the sections that minimally disrupt overall coherence.
>
> Respond in the following JSON format:
>
> {"Revision Suggestion": "specific suggestion"}

Document-Level Reviser

> Below is an electronic medical record with flagged issues:
>
> [record]
>
> Please revise the record according to the following suggestion: [feedback].
>
> Return the updated record in JSON format, including all sections. The content of each section should be a single string.

Corpus-Level Agents: For the corpus-level stage, we use statistical analysis tools as the critic and adviser rather than LLMs. Therefore, no natural language instructions are required for these agents; their operations are fully automated and operate on dataset-wide distributions. For the corpus-level reviser, each sample in the selected subset is modified individually, using the same type of instructions as the document-level reviser.

E.2 LLM BACKBONES

We use the following pretrained large language models in our experiments:

- **Qwen2.5** (Yang et al., 2025): Licensed under the Apache 2.0 License[1]. We use the model checkpoints available on Huggingface[2].
- **LLaMA 3.1** (Dubey et al., 2024): Licensed under the LLaMA 3.1 Community License[3]. We use the model checkpoints available on Huggingface[4].
- **Meditron 3** (Chen et al., 2023): Licensed under the LLaMA 3.1 Community License[5]. We use the model checkpoints available on Huggingface[6].
- **DeepSeek-R1-Distill-Llama** (Guo et al., 2025): Licensed under the MIT License[7]. We use the model checkpoints available on Huggingface[8].

[1] https://huggingface.co/Qwen/Qwen2.5-7B-Instruct/blob/main/LICENSE
[2] https://huggingface.co/Qwen
[3] https://www.llama.com/llama3_1/license/
[4] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
[5] https://www.llama.com/llama3_1/license/
[6] https://huggingface.co/OpenMeditron/Meditron3-8B
[7] https://github.com/deepseek-ai/DeepSeek-R1/blob/main/LICENSE
[8] https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-8B

文档级评论者

> 以下是由多个部分组成的电子病历：[记录]
>
> 请评估该记录是否满足以下要求： [要求]。识别各部分之间的任何冲突或不合理的陈述。
> 请使用以下 JSON 格式回应：{“符
> 合要求”： true/false}

文档级顾问

> 以下记录存在违反要求的问题： [要求]:[记录]请提供有针对性的建议以解决该问题。优先考虑对整体连贯性影响最小的部分进行修改。请按以下 JSON 格式回复：{“修订建议”：”具体建议”}

文档级修订器

> 以下是一份带有标记问题的电子病历：[记录]请根据以下建议修改病历： [反馈]。请以 JSON 格式返回更新后的病历，包括所有部分。每个部分的内容应为单一字符串。

语料库级别代理：对于语料库级别阶段，我们使用统计分析工具作为批评者和顾问，而不是大型语言模型。因此，这些代理不需要自然语言指令；它们的操作是完全自动化的，并在整个数据集的分布上运行。对于语料库级别的修订者，所选子集中的每个样本都会被单独修改，使用与文档级修订者相同类型的指令。

E.2 大型语言模型骨干

我们在实验中使用了以下预训练的大型语言模型：

- Qwen2.5（Yang 等，2025）： 根据 Apache 2.0 许可证获得授权[1]。我们使用 Huggingface 上提供的模型检查点[2]。
- LLaMA 3.1（Dubey 等，2024）： 根据 LLaMA 3.1 社区许可授权[3]。我们使用 Huggingface 上可用的模型检查点[4]。
- Meditron 3（Chen 等，2023）： 根据 LLaMA 3.1 社区许可证获得授权[5]。我们使用 Huggingface 上可用的模型检查点[6]。
- DeepSeek-R1-Distill-Llama (Guo 等, 2025)：根据 MIT 许可证授权[7]。我们使用 Huggingface 上提供的模型检查点[8]。

[1] https://huggingface.co/Qwen/Qwen2.5-7B-Instruct/blob/main/LICENSE[2]
https://huggingface.co/Qwen[3] https://www.llama.com/llama3_1/license/[4]
https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct[5]
https://www.llama.com/llama3_1/license/[6]
https://huggingface.co/OpenMeditron/Meditron3-8B[7]
https://github.com/deepseek-ai/DeepSeek-R1/blob/main/LICENSE[8]
https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-8B

E.3 HYPERPARAMETERS

For LLM-CARe and LLM Direct—we adopt the default generation configuration provided with each model checkpoint. For all refinement stages (corpus, section, and document), the Critic, Adviser, and Reviser agents are iterated for two cycles before proceeding to the next stage. We empirically observed that additional iterations beyond two provided negligible improvements in EMR quality and downstream task performance.

For EMR quality evaluation, we use greedy decoding to ensure deterministic outputs. For downstream tasks, we fine-tune Qwen2.5-0.5B-Instruct using the AdamW optimizer with a batch size of 16, a learning rate of 2e-5, and a cosine learning rate scheduler with 5% warmup steps. The model is fine-tuned for 3 epochs on the synthetic dataset and evaluated on the real test set.

E.4 SOFTWARE

EMR generation with LLMs is conducted using vLLM (Kwon et al., 2023). PyTorch (Paszke et al., 2019) is used for training and inference of non-LLM baselines. Fine-tuning of downstream task models is performed using the Huggingface Transformers library (Wolf et al., 2020). Evaluations are also executed using vLLM.

E.5 COMPUTATIONAL RESOURCES

All experiments—except for EMR quality evaluation—are conducted on NVIDIA RTX 4090 GPU with 24GB of memory. EMR quality evaluation, which uses Qwen2.5-32B-Instruct, is performed on NVIDIA A100 GPU with 80GB of memory.

F ROBUSTNESS OF LLM-AS-JUDGE EVALUATION

Table 10: Quality scores evaluated by different LLMs.

Evaluation LLM	Method	Content Completeness	Medical Correctness	Context Consistency
Qwen2.5-32B	LSTM	70.8	65.0	21.7
	mtGAN	55.8	51.8	21.4
	MedSyn	84.8	95.3	91.9
	LLM Direct	77.1	90.7	87.9
	LLM-CARe (ours)	**91.2**	**98.6**	**93.8**
GPT-OSS-20B	LSTM	73.9	88.9	8.9
	mtGAN	62.8	87.7	13.4
	MedSyn	79.9	97.3	86.9
	LLM Direct	81.1	96.7	70.2
	LLM-CARe (ours)	**93.4**	**99.0**	**91.3**
Deepseek-V3.2-Exp	LSTM	69.9	72.5	2.6
	mtGAN	57.9	64.0	0.7
	MedSyn	80.6	97.5	85.0
	LLM Direct	79.9	96.8	77.8
	LLM-CARe (ours)	**91.1**	**98.5**	**89.8**

To examine whether our intrinsic quality evaluation is robust to the choice of LLM judge, we performed an extended cross-model assessment using three independent evaluators: Qwen2.5-32B, GPT-OSS-20B (OpenAI, 2025), and DeepSeek-V3.2-Exp (DeepSeek-AI, 2025). Qwen2.5-32B is our original judge; GPT-OSS-20B was included due to its reported strength in medical-language understanding tasks, and DeepSeek-V3.2-Exp (API) was added as a large commercial model to further test evaluation generality across architectures and training regimes. Owing to the computational cost of the API, DeepSeek evaluation was conducted on 10% sampled data.

E.3 超参数

对于 LLM-CARe 和 LLM Direct——我们采用每个模型检查点提供的默认生成配置。对于所有的精炼阶段（语料、章节和文档），批评者（Critic）、顾问（Adviser）和修订者（Reviser）代理在进入下一个阶段之前会迭代两轮。我们通过经验观察发现，超过两轮的额外迭代对 EMR 质量和下游任务性能的提升微乎其微。

对于电子病历质量评估，我们使用贪婪解码以确保输出的确定性。对于下游任务，我们使用 AdamW 优化器对 Qwen2.5-0.5B-Instruct 进行微调，批量大小为 16，学习率为 2e-5，并使用带有 5% 预热步骤的余弦学习率调度器。该模型在合成数据集上微调 3 个周期，并在真实测试集上进行评估。

E.4 软件

使用 vLLM (Kwon 等, 2023) 生成含有 LLM 的电子病历。PyTorch (Paszke 等, 2019) 被用于非 LLM 基线模型的训练和推理。下游任务模型的微调是使用 Huggingface Transformers 库 (Wolf 等, 2020) 进行的。评估也使用 vLLM 执行。

E.5 计算资源

除了 EMR 质量评估之外，所有实验都在配备 24GB 内存的 NVIDIA RTX 4090 GPU 上进行。使用 Qwen2.5-32B-Instruct 的 EMR 质量评估在配备 80GB 内存的 NVIDIA A100 GPU 上进行。

LLM作为评审的稳健性

表10：由不同大型语言模型评估的质量分数。

评估大型语言模型	方法	内容完整性	医疗正确性	上下文一致性
Qwen2.5-32B	LSTM	70.8	65.0	21.7
	mtGAN	55.8	51.8	21.4
	医合	84.8	95.3	91.9
	大型语言模型直连	77.1	90.7	87.9
	LLM-CARe（我们的）	**91.2**	**98.6**	**93.8**
GPT-OSS-20B	LSTM	73.9	88.9	8.9
	mtGAN	62.8	87.7	13.4
	医合	79.9	97.3	86.9
	大型语言模型直连	81.1	96.7	70.2
	LLM-CARe（我们的）	**93.4**	**99.0**	**91.3**
Deepseek-V3.2-Exp	LSTM	69.9	72.5	2.6
	mtGAN	57.9	64.0	0.7
	医合	80.6	97.5	85.0
	大型语言模型直连	79.9	96.8	77.8
	LLM-CARe（我们的）	**91.1**	**98.5**	**89.8**

为了检验我们的内在质量评估是否对 LLM 评审的选择具有鲁棒性，我们进行了扩展的跨模型评估，使用了三位独立评审员：Qwen2.5-32B、GPT-OSS-20B（OpenAI，2025）和 DeepSeek-V3.2-Exp（DeepSeek-AI，2025）。Qwen2.5-32B 是我们的原始评审员；GPT-OSS-20B 被纳入是因为其在医学语言理解任务中表现出的优势，而 DeepSeek-V3.2-Exp（API）作为一个大型商业模型被加入，以进一步测试评估在不同架构和训练体系下的普遍性。由于 API 的计算成本，DeepSeek 的评估仅在 10% 的抽样数据上进行。

All three evaluators were provided with the same scoring instructions used in our main experiments. As shown in Table 10, the absolute scores vary across models—as expected due to differences in calibration and alignment—but the overall ranking pattern remains consistent, with LLM-CARe achieving the top scores across all intrinsic quality dimensions under all judges. The baselines also maintain broadly similar relative positions, with only minor ordering differences between closely performing systems. These results indicate that our evaluation conclusions are not tied to a specific model family and that LLM-CARe demonstrates robust superiority under multiple independent evaluators.

G Subgroup Analysis Across Demographic and Disease Frequency

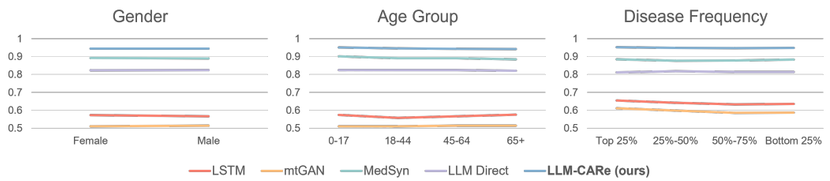


Figure 12: Subgroup analysis of intrinsic EMR quality.

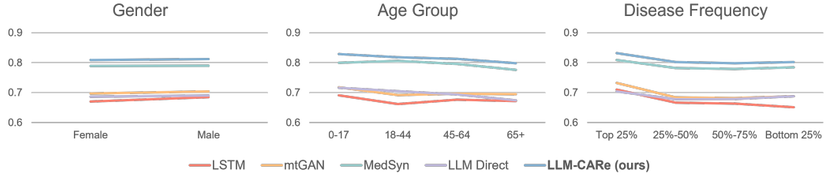


Figure 13: Subgroup analysis of downstream task performance.

To examine whether LLM-CARe amplifies or mitigates subgroup disparities, we conducted a stratified evaluation across three grouping dimensions: gender, age (0–17, 18–44, 45–64, 65+), and disease-frequency strata. Figure 12 reports intrinsic EMR quality across subgroups, and Figure 13 shows downstream performance under the same partitions. LLM-CARe consistently achieves the highest scores across all subgroups, and the variation across demographic and frequency groups is comparable to or smaller than that of the baselines. These results indicate that LLM-CARe does not reinforce demographic or clinical biases and maintains robustness across diverse subpopulations, supporting its applicability to balanced synthetic EMR corpus construction.

H Full Results of EMR Quality

Figure 14 provides the complete breakdown of EMR quality across all evaluated criteria, extending the representative results presented in Figure 5 of the main text. These detailed results offer a more comprehensive view of how different methods perform with respect to each quality dimension.

I Privacy Evaluation via Membership Inference Attack

To assess whether synthetic EMRs generated by different methods inadvertently reveal information from real patient records, we conduct a membership inference attack (MIA)—a standard privacy evaluation technique that tests whether an attacker can distinguish training examples from non-

三位评估者都使用了我们主要实验中相同的评分说明。如表10所示，绝对分数因模型而异——这是预期的，因为校准和对齐存在差异——但整体排名模式保持一致，LLM-CARe在所有评估者下所有内在质量维度中均取得最高分。基线模型的相对位置也大体相似，仅在表现接近的系统之间存在轻微的排序差异。这些结果表明，我们的评估结论并不依赖于特定的模型家族，并且LLM-CARe在多个独立评估者下展示了稳健的优势。

G 子组分析跨人口统计和疾病频率

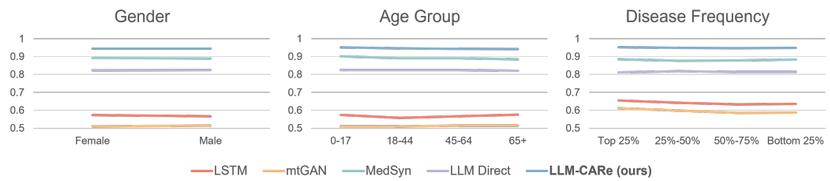


图12：内在电子病历质量的亚组分析。

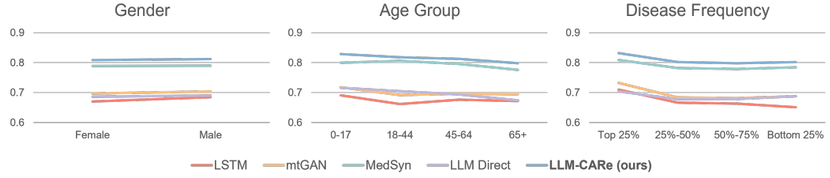


图13：下游任务表现的子组分析。

为了检验 LLM-CARe 是否会放大或减轻子群差异，我们在三个分组维度上进行了分层评估：性别、年龄（0–17，18–44，45–64， 65+）以及疾病频率层级。图 12 报告了各子群的内在电子病历（EMR）质量，图 13 显示了在相同分组下的下游性能。LLM-CARe 在所有子群中始终获得最高分，其在不同人口统计学和频率群体间的变化与基线相比相当或更小。这些结果表明，LLM-CARe 不会强化人口统计学或临床偏差，并能在多样化子群中保持稳健性，支持其在平衡的合成电子病历语料构建中的适用性。

H EMR 质量的完整结果

图14提供了所有评估标准下电子病历（EMR）质量的完整细分，扩展了主文中图5所呈现的代表性结果。这些详细结果提供了关于不同方法在各质量维度上的表现的更全面的视角。

我通过成员推断攻击进行隐私评估

为了评估由不同方法生成的合成电子病历是否无意中泄露了真实患者记录的信息，我们进行了一次成员推断攻击（MIA）——一种标准的隐私评估技术，用于测试攻击者是否能够区分训练样本和非训练样本。

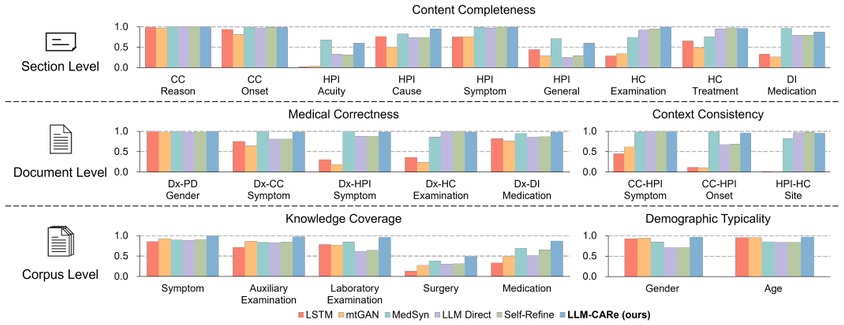


Figure 14: Fine-Grained EMR quality evaluation across three level of criteria. Abbreviations of EMR sections: CC-Chief Complaint, HPI-History of Present Illness, HC-Hospital Course, DI-Discharge Instructions, Dx-Diagnosis.

members. An accuracy of 0.50 corresponds to random guessing; values close to 0.50 therefore indicate stronger privacy, as the attacker cannot reliably infer membership.

Table 11 reports the attack accuracy across all methods. LLM-CARe and LLM Direct both achieve values near 0.50, as expected, since neither method accesses real EMR text during generation. LSTM and mtGAN also remain close to 0.50. Although these models are trained on real EMRs, their limited model capacity—combined with the complexity of multi-section EMRs used in our setting—reduces their ability to memorize full clinical notes. In contrast, MedSyn, which directly uses real EMRs as in-context exemplars, exhibits higher attack accuracy, indicating elevated privacy risk due to closer exposure to specific real records.

Table 11: Membership inference attack accuracy (closer to 0.50 is better).

Method	Attack Accuracy
LSTM	0.499
mtGAN	0.504
MedSyn	0.533
LLM Direct	0.500
LLM-CARe (ours)	0.504

These results show that LLM-CARe introduces no observable privacy risk and that its generation process remains indistinguishable from non-member data, consistent with the fact that it operates without using any real EMR text.

J Additional Validation of Downstream Task Reliability

To further examine the robustness of downstream task results, we conducted two complementary evaluations.

Table 12: Downstream task performance using Llama-3.2-1B-Instruct as the backbone model.

Method	Diagnosis Prediction		Examination Recommendation		Treatment Recommendation	
	Micro	Macro	Micro	Macro	Micro	Macro
LSTM	68.5	67.1	74.7	75.4	55.6	48.6
mtGAN	78.8	77.7	72.8	73.6	58.9	53.1
MedSyn	77.2	77.8	82.4	81.7	72.0	68.7
LLM Direct	78.9	79.0	61.9	62.8	59.4	57.0
LLM-CARe (ours)	**80.1**	**79.8**	**83.3**	**83.2**	**74.7**	**71.8**

First, we replaced the backbone model used for downstream classifiers and retrained all methods using Llama-3.2-1B-Instruct. As reported in Table 12, the relative ordering of all methods remained

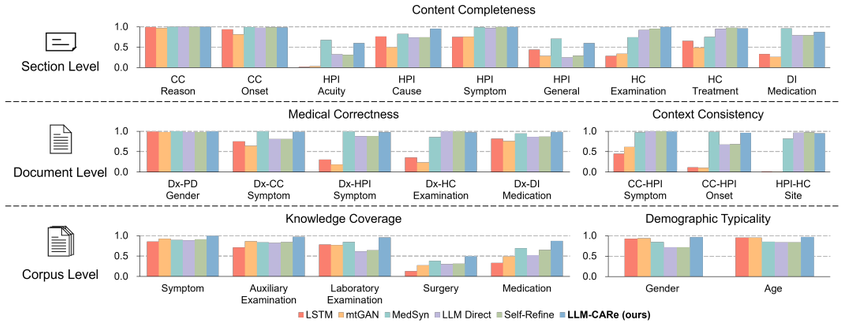


图14：跨三个层次标准的电子病历精细质量评估。电子病历部分缩写：CC-主诉，HPI-现病史，HC-住院过程，DI-出院指导，Dx-诊断。

成员。0.50 的准确率对应于随机猜测；因此接近 0.50 的值表示更强的隐私性，因为攻击者无法可靠地推断成员身份。

表 11 报告了所有方法的攻击准确率。LLM-CARe 和 LLM Direct 都达到了接近 0.50 的数值，这是预期的，因为两种方法在生成过程中都没有访问真实的电子病历（EMR）文本。LSTM 和 mtGAN 也保持在接近 0.50 的水平。尽管这些模型是在真实 EMRs 上训练的，但它们有限的模型容量——再加上我们设置中使用的多部分 EMRs 的复杂性——降低了它们记忆完整临床记录的能力。相比之下，MedSyn 直接使用真实 EMRs 作为上下文示例，表现出更高的攻击准确率，

表 11：成员推断攻击准确率（越接近 0.50 越好）。

方法	攻击准确性
LSTM	0.499
mtGAN	0.504
医合 大型语言模型直连	0.533
LLM-CARe（我们的）	0.500
	0.504

由于更接近特定真实记录而导致隐私风险升高。

这些结果表明，LLM-CARe 不会引入可观察的隐私风险，其生成过程与非成员数据无法区分，这与其在操作时不使用任何真实 EMR 文本的事实一致。

J 下游任务可靠性的额外验证

为了进一步检验下游任务结果的稳健性，我们进行了两项互补评估。

表12：使用Llama-3.2-1B-Instruct作为骨干模型的下游任务性能。

方法	诊断预测		考试推荐		治疗推荐	
	微观	宏观	微观	宏	微	宏
LSTM	68.5	67.1	74.7	75.4	55.6	48.6
mtGAN	78.8	77.7	72.8	73.6	58.9	53.1
医合 大型语言模型直连	77.2	77.8	82.4	81.7	72.0	68.7
LLM-CARe（我们的）	78.9	79.0	61.9	62.8	59.4	57.0
	80.1	**79.8**	**83.3**	**83.2**	**74.7**	**71.8**

首先，我们替换了用于下游分类器的骨干模型，并使用 Llama-3.2-1B-Instruct 重新训练了所有方法。如表 12 所示，所有方法的相对排序保持不变

stable, with LLM-CARe achieving the highest accuracy on diagnosis prediction, examination recommendation, and treatment recommendation. This consistency under a different model architecture indicates that performance gains do not depend on properties of the Qwen family, but instead arise from the proposed multi-level refinement framework.

Table 13: Diagnosis prediction accuracy on MIMIC-IV-Note dataset.

Method	Micro Average	Macro Average
LSTM	0.587	0.552
mtGAN	0.589	0.555
MedSyn	0.584	0.571
LLM Direct	0.589	0.532
LLM-CARe (ours)	**0.606**	**0.590**

Second, to test whether models benefit from superficial alignment with the textual format of the generation prompts, we constructed an additional diagnosis-prediction task using MIMIC-IV-Note. We selected only records corresponding to the same disease categories but originating from a different clinical institution and written in a distinctly different narrative style. All models trained on synthetic corpora exhibit lower accuracy due to the distributional and stylistic shift, yet LLM-CARe again achieves the best performance (Table 13), demonstrating that the downstream advantages are not tied to prompt-format similarity but to improved clinical quality in the synthesized EMRs.

K Robustness to Prompting Styles and Initial Draft Quality

Table 14: Intrinsic EMR quality under prompt rephrasing and improved initial drafts.

Method	Section Level	Document Level		Corpus Level	
	Content Completeness	Medical Correctness	Context Consistency	Demographic Typicality	Knowledge Coverage
LLM-CARe (ours)	91.2	98.6	93.8	96.8	94.1
+Rephrased Prompts	94.1	98.0	93.8	96.8	95.1
+Better Initial Draft	93.3	98.6	94.0	96.8	96.2

Table 15: Downstream prediction performance (micro/macro accuracy) under prompt and draft perturbations.

Method	Diagnosis Prediction		Examination Recommendation		Treatment Recommendation	
	Micro	Macro	Micro	Macro	Micro	Macro
LLM-CARe (ours)	82.6	82.4	85.3	85.2	76.9	74.1
+Rephrased Prompts	81.1	80.7	83.4	83.4	75.3	73.0
+Better Initial Draft	82.6	81.8	84.7	84.4	76.8	73.9

We evaluated the stability of LLM-CARe under variations in prompting style and initial draft quality. First, all agent prompts were substantially rephrased using GPT-5 while preserving only high-level intent. Second, to test sensitivity to draft quality, we replaced the LLM-Direct drafts with higher-quality MedSyn drafts. Tables 14 and 15 report intrinsic quality and downstream task performance.

Across both perturbations, numerical differences are small and the overall ranking of methods remains unchanged. This robustness reflects the structure of LLM-CARe: quality requirements are decomposed into explicit criteria, and each agent operates on a single criterion at a time, making the refinement cycle insensitive to prompt wording. Likewise, the staged refinement progressively corrects deficiencies from multiple dimensions, reducing dependence on the initial draft. These properties together ensure that moderate changes in prompt phrasing or draft quality do not materially affect the final refined EMRs.

稳定，其中LLM-CARe在诊断预测、检查推荐和治疗推荐上取得了最高的准确率。这种在不同模型架构下的一致性表明，性能提升并不依赖于Qwen系列的特性，而是来源于所提出的多层次改进框架。

表 13：MIMIC-IV-Note 数据集上的诊断预测准确率。

方法	微平均	宏平均
LSTM	0.587	0.552
mtGAN	0.589	0.555
医合	0.584	0.571
大型语言模型直连	0.589	0.532
LLM-CARe（我们的）	**0.606**	**0.590**

其次，为了测试模型是否能从与生成提示文本格式的表面对齐中受益，我们使用 MIMIC-IV-Note 构建了一个额外的诊断预测任务。我们只选择对应相同疾病类别的记录，但这些记录来自不同的临床机构，并且以明显不同的叙述风格编写。由于分布和风格的变化，所有在合成语料上训练的模型的准确率都较低，但 LLM-CARe 再次取得了最佳表现（表 13），这表明下游优势并不依赖于提示格式的相似性，而是源于合成电子病历中临床质量的提升。

对提示风格和初稿质量的鲁棒性

表14：在提示重述和改进初始草稿下的内在EMR质量。

方法	章节级别	文档级		语料库级别	
	内容完整性	医疗正确性	上下文一致性	人口统计典型性	知识报道
LLM-CARe（我们的）	91.2	98.6	93.8	96.8	94.1
+重写的提示	94.1	98.0	93.8	96.8	95.1
+更好的初稿	93.3	98.6	94.0	96.8	96.2

表15：在提示和草稿扰动下的下游预测性能（微观/宏观准确率）。

方法	诊断预测		考试推荐		治疗推荐	
	微观	宏观	微观	宏	微	宏
LLM-CARe（我们的）	82.6	82.4	85.3	85.2	76.9	74.1
+重写的提示	81.1	80.7	83.4	83.4	75.3	73.0
+更好的初稿	82.6	81.8	84.7	84.4	76.8	73.9

我们评估了 LLM-CARe 在提示风格和初始草稿质量变化下的稳定性。首先，所有代理提示都使用 GPT-5 进行了大幅度改写，仅保留高级意图。其次，为了测试对草稿质量的敏感性，我们将 LLM-Direct 草稿替换为质量更高的 MedSyn 草稿。表 14 和表 15 报告了内在质量和下游任务表现。

在这两种扰动下，数值差异很小，方法的整体排名保持不变。这种稳健性反映了 LLM-CARe 的结构：质量要求被分解为明确的标准，每个智能体一次只处理一个标准，使得优化循环对提示措辞不敏感。同样，分阶段优化会从多个维度逐步纠正缺陷，减少对初稿的依赖。这些特性共同确保了提示措辞或初稿质量的适度变化不会实质性地影响最终优化的电子病历。

L Details of Clinician Evaluation

L.1 Reliability of Clinician Evaluation

Table 16: Agreement between clinicians and LLM-based evaluation across different sample sizes.

# Samples	Clinician-LLM Agreement		Inter-Clinician Agreement	
	Cohen's Kappa	Confidence Interval (95%)	Fleiss's Kappa	Confidence Interval (95%)
100	0.837	[0.804, 0.868]	0.947	[0.928, 0.964]
150	0.842	[0.816, 0.868]	0.948	[0.933, 0.962]
200	0.833	[0.809, 0.855]	0.950	[0.937, 0.962]

To ensure that clinician evaluation reflects a broad range of clinical scenarios, all clinician-labeled samples were obtained through stratified sampling. Diseases were grouped by frequency, and each group contributed proportionally to the evaluation set. The sampling was likewise balanced across generation methods, gender, and age groups, ensuring that the annotated EMRs covered both common and less-common situations and represented diverse demographic and clinical patterns.

To examine the stability of human evaluation, we increased the evaluation size from 100 to 150 and then to 200 EMRs. As shown in Table 16, clinician–LLM agreement (Cohen's Kappa) remains consistently above 0.8 across all three subset sizes, and inter-clinician agreement (Fleiss's Kappa) stays above 0.9. Confidence intervals narrow as sample size increases, but the agreement values themselves remain highly similar, indicating that both human annotations and LLM-based evaluations are stable and reliable.

L.2 Examples of Disagreement Between LLM and Clinicians

We present examples of disagreement observed in our human evaluation.

LLM–Clinician Disagreement
Criterion: Discharge instructions should specify both the medication name and its dosage/usage.
Discharge Instruction: Continue oral antibiotic therapy...
LLM: Correct
Clinician: Wrong

In such cases, the LLM treats non-specific statements (e.g., "continue oral antibiotic therapy") as sufficiently informative, whereas clinicians regard them as incomplete because essential details such as drug name and dosage are missing. This reflects the LLM's more permissive interpretation of medication-related content.

Inter–Clinician Disagreement
Criterion: The history of present illness should describe general conditions (diet, sleep, mental status, bowel/urination, weight change).
History of Present Illness: The patient developed dizziness and fatigue two days ago... Appetite and sleep are normal; bowel and urination regular; no weight change.
Clinician A: Correct
Clinician B: Wrong

Here, clinicians differ in how they interpret the requirement to document mental status. Some consider dizziness and fatigue to implicitly convey reduced alertness, whereas others expect an explicit statement. Such differences reflect variation in documentation style rather than clinical competence and represent a common source of mild disagreement in narrative EMR review.

临床医生评估详细信息

L.1 临床评估的可靠性

Tab第16条：临床医生与基于大语言模型的评估在不同样本量下的一致性 .

# 样本	临床医生-大语言模型一致性		临床医生间一致性	
	科恩 ’s 卡帕	自信区间（95%）	弗莱斯 ’s 卡帕	自信区间（95%）
100	0.837	[0.804, 0.868]	0.947	[0.928, 0.964]
150	0.842	[0.816, 0.868]	0.948	[0.933, 0.962]
200	0.833	[0.809, 0.855]	0.950	[0.937, 0.962]

为了确保临床医生的评估能够反映广泛的临床情景，所有由临床医生标注的样本都是通过分层抽样获得的。疾病按照发生频率分组，每个组按比例贡献到评估集。抽样在生成方法、性别和年龄组之间同样保持平衡，确保标注的电子病历涵盖常见和较少见的情况，并代表多样的人口统计学和临床模式。

为了检验人工评估的稳定性，我们将评估样本量从100份增加到150份，再增加到200份病历。如表16所示，临床医生与大型语言模型（Cohen’s Kappa）的一致性在三个子集规模中始终保持在0.8以上，且临床医生之间的一致性（Fleiss’s Kappa）保持在0.9以上。随着样本量的增加，置信区间变窄，但一致性数值本身保持高度相似，这表明人工标注和基于大型语言模型的评估均稳定且可靠。

L.2 大型语言模型与临床医生之间分歧的例子

我们展示了在人工评估中观察到的分歧例子。

大型语言模型–临床医生分歧
标准：出院指导应明确说明药物名称及其剂量/用法。出院指导：继续口服抗生素治疗…… 语言模型：正确 医生：错误

在这种情况下，大型语言模型（LLM）将非具体陈述（例如，“继续口服抗生素治疗”）视为足够信息，而临床医生则认为这些陈述不完整，因为缺少药物名称和剂量等关键信息。这反映了LLM对与药物相关内容更宽容的解读。

临床医生间的分歧
标准：现病史应描述一般情况（饮食、睡眠、精神状态、排便/排尿、体重变化）。现病史：患者两天前出现头晕和乏力……食欲和睡眠正常；排便和排尿规律；体重无变化。医师A：正确 医师B：错误

在这里，临床医生在如何解释记录心理状态的要求上存在差异。有些人认为头晕和疲劳可以隐含地传达警觉性降低，而其他人则期望有明确的说明。这种差异反映的是记录风格的不同，而不是临床能力的差异，并且是叙述性电子病历审查中常见的小分歧来源。

L.3 Evaluation Schema

To verify the reliability of our evaluation, we asked licensed clinicians to assess the quality of synthetic EMRs. Clinicians were instructed to read each synthetic record carefully and then answer several yes/no questions regarding **completeness**, **consistency**, and **correctness**. The questions were designed to be simple binary judgments to ensure reproducibility. The detailed labeling instructions are as follows:

> Please review the synthetic EMR text shown below.
>
> **Synthetic EMR:**
> **Gender:** Male
> **Age:** 45 years old
> **Primary Diagnosis:** Pneumonia
> **Chief Complaint:** Fever and cough for 3 days ...
> **History of Present Illness:** Patient developed fever three days ago, accompanied by cough and mild chest pain ...
> ...
>
> Based on the above synthetic EMR, please answer the following questions. For each question, mark your judgment in the blank: `Yes` if the requirement is satisfied, `No` otherwise.
> **Completeness**
> - Does the history of present illness include major symptoms? Answer: ____ (`Yes/No`)
> - ...
>
> **Consistency**
> - Are the symptoms in the chief complaint consistent with those in the history of present illness? Answer: ____ (`Yes/No`)
> - ...
>
> **Correctness**
> - Is the patient's sex valid given the diagnosis? Answer: ____ (`Yes/No`)
> - ...

Confidence interval estimation: To quantify agreement, we report Cohen's Kappa and Fleiss's Kappa with 95% confidence intervals. The intervals were computed using a non-parametric bootstrap procedure with 10,000 resamples, which provides uncertainty estimates without assuming normality of the statistics.

M Computational Cost

On a single RTX 4090 GPU, producing 38k synthetic EMRs takes approximately 13 hours with direct generation, whereas LLM-CARe requires about 36 hours, which includes initial draft generation and all three refinement stages. Although the framework performs three levels of Critic–Adviser–Reviser interactions, the actual overhead is moderated by two factors. First, the outputs of these agents are much shorter than full EMRs, making each refinement step relatively lightweight. Second, if a draft already satisfies the criteria at a given stage, it bypasses subsequent agents, avoiding unnecessary iterations. Moreover, the computational cost scales linearly with the number of EMRs, and the cyclic refinement process can be parallelized across disease categories, making the framework feasible for scaling to larger corpora.

N Use of LLMs

In preparing this manuscript, we used LLM solely as an assistive tool for text refinement, including grammar correction, and language polishing. The research ideas, experimental design, implementation, and analysis were entirely conceived and executed by the authors.

L.3 评估方案

为了验证我们评估的可靠性，我们请持牌临床医生评估合成电子病历的质量。临床医生被要求仔细阅读每一份合成记录，然后回答几个关于完整性、一致性和正确性的是/否问题。这些问题被设计为简单的二元判断，以确保可重复性。详细的标注说明如下：

> P请审查下方显示的合成电子病历文本 low.
>
> 合成电子病历： 性别：男性 年龄：45岁 主要诊断：肺炎 主诉：发热和咳嗽3天 病史：患者三天前出现发热，伴有咳嗽和轻微胸痛。
>
> ...
>
> 根据上述综合电子病历，请回答以下问题。对于每个问题，在空白处标注您的判断：如果满足要求，请标记“是”，否则标记“否”。
>
> **完整性**
> • 现病史包括主要症状吗？回答：（是/否） ____
> •...
>
> **一致性**
> • 主诉中的症状与现病史中的症状是否一致？回答：（是/否）
> ____
> •...
>
> **正确性**
> • 根据诊断，患者的性别是否有效？ 答：（是/否） ____
> •...

置信区间估计：为了量化一致性，我们报告 Cohen's Kappa 和 Fleiss's Kappa 的 95% 置信区间。这些区间是使用非参数自助法（bootstrap）程序进行计算的，重采样 10,000 次，该方法在不假设统计量服从正态分布的情况下提供不确定性估计。

M 计算成本

在单个 RTX 4090 GPU 上，直接生成 38k 合成电子病历大约需要 13 小时，而使用 LLM-CARe 则需要约 36 小时，这包括初稿生成和所有三个优化阶段。虽然该框架执行了三个层次的评论者–顾问–修订者交互，但实际开销受两个因素的调节。首先，这些代理的输出远比完整的电子病历短，使得每个优化步骤相对轻量。其次，如果某个草稿在特定阶段已经满足标准，它将跳过后续代理，避免不必要的迭代。此外，计算成本会随电子病历数量线性增加，并且循环优化过程可以在不同疾病类别间并行化，使该框架在扩展到更大语料库时仍然可行。

LLMS 的使用

在准备本手稿时，我们仅将大型语言模型（LLM）用作文本润色的辅助工具，包括语法纠正和语言修饰。研究理念、实验设计、实施和分析完全由作者独立构思和执行。